

Senior High School



Lesson Exemplar in General Mathematics

Quarter 4

Unit

12

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Lesson Exemplar for General Mathematics
Quarter 4: Unit 12

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Republic of the Philippines
Department of Education
 BUREAU OF LEARNING DELIVERY

LESSON EXEMPLAR			
Learning Area:	GENERAL MATHEMATICS	Grade Level:	11
Semester:	SECOND	Quarter:	FOURTH
		Unit:	12

I. LEARNING GOALS	
Content Standard	The learners demonstrate knowledge and understanding of logical propositions, syllogisms, and fallacies.
Performance Standard	By the end of the quarter, learners evaluate arguments using logic in real-world contexts (e.g., critically analyzing misleading arguments or statements in advertisements, politics, or social media) through Role-Playing Real-World Arguments.
Learning Competencies	The learners shall... 1. identify logical propositions; 2. differentiate simple from compound propositions; 3. illustrate basic logical operations; 4. illustrate the different forms of conditional propositions; 5. illustrate different types of tautologies and contradictions using practical situations; and 6. determine the truth values of propositions in a practical situation.
II. REFERENCES and MATERIALS	Textbook and Modules Department of Education. (2021). <i>General Mathematics Quarter 2 – Module 13: Propositions (First Edition)</i> [Self-Learning Module]. Department of Education. (2021). <i>General Mathematics Quarter 2 – Module 14: Simple and Compound Propositions (First Edition)</i> [Self-Learning Module]. Department of Education. (2021). <i>General Mathematics Quarter 2 – Module 15: Logical Operators (First Edition)</i> [Self-Learning Module]. Department of Education. (2021). <i>General Mathematics Quarter 2 – Module 16: Truth Values of Propositions (v2)</i> [Self-Learning Module].

Department of Education. (2021). *General Mathematics Quarter 2 – Module 17: Logical Equivalence and Conditional Propositions (v2)*

Department of Education. (2021). *General Mathematics Quarter 2 – Module 18: Tautologies and Fallacies (v2)* [Self-Learning Module].

Department of Education. (2021). *General Mathematics Quarter 2 – Module 19: Validity of Categorical Syllogisms (v2)* [Self-Learning Module].

Department of Education. (2021). *General Mathematics Quarter 2 – Module 20: Valid Arguments and Fallacies (v2)* [Self-Learning Module].

Materials

1. Whiteboard and markers
2. Projector
3. Handouts or sample sentences
4. Sticky notes
5. Timer
6. Pre-made sentence cards
7. Clipboards or desk space
8. Visual cue cards
9. True/False placards (or bond paper for learners to make their own)
10. Printed list of declarative statements
11. Learner notebooks
12. Highlighters or colored pens
13. Sorting charts (One Idea / More than One Idea)
14. Index cards
15. “Clue word” chart (for logical connectives)
16. Activity sheets (with propositions, operators, truth tables, etc.)
17. Manila paper/cartolina

	18. Markers	
III. CONTENT	Lesson 12.1: Logical Propositions Recommended time allotment: 2 hours	
IV. OBJECTIVES	By the end of the lesson, the learners... 1. define logical propositions; 2. illustrate logical proposition; 3. recognize whether a statement is a proposition or not; 4. define the two types of propositions in logic; 5. distinguish between simple and compound propositions; 6. define the fundamental logical operators—conjunction, disjunction, negation, conditional, and biconditional; 7. identify the logical operator(s) present in a given proposition; 8. apply various logical operations to propositions including constructing truth tables; and 9. evaluate arguments using principles of logical propositions and logical connections in real-life contexts.	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1: Sentence Types Recall This activity helps you to connect your prior knowledge of sentence types from English to the new concept of logical propositions in Mathematics. By recalling declarative, interrogative, imperative, and exclamatory sentences, you can recognize that only declarative sentences that can be judged as true or false qualify as logical propositions. This prepares you to define and understand propositions more clearly. Instructions: 1. Construct your own examples of the four types of sentences listed below. Write one sentence for each type in your notebook or on the worksheet provided: a. <i>Declarative</i> – states a fact or opinion b. <i>Interrogative</i> – asks a question	ActivityA.1: Sentence Types Recall This activity activates learners’ prior knowledge of English grammar (sentence types) and connects it to the new concept of logical propositions in Mathematics. It helps learners realize that only declarative sentences that can be evaluated as true or false qualify as propositions. Strategy 1: Class Brainstorming Purpose: To activate students’ prior knowledge of sentence types and guide them in discovering, through their own examples, which types can be judged as true or false (i.e., propositions). Materials: ✓ Whiteboard/marker or projector ✓ paper or index cards, sticky notes (optional) ✓ timer

	c. <i>Imperative</i> – gives a command or request d. <i>Exclamatory</i> – shows strong emotion 2. Exchange your work with a classmate or groupmate. Identify and label the type of each sentence your classmate has written. 3. Analyze your sentences: ✓ Which of these sentences can be judged as <i>true</i> or <i>false</i>? ✓ Which cannot be judged as either? <i>Example:</i> ✓ “The earth has only one moon.” → Can be judged true. ✓ “What time is it?” → Cannot be judged true or false. 4. Discuss with your group or class: a. Which sentence types can be considered logical propositions in Mathematics? b. Why do only declarative sentences qualify as propositions? Process Questions: 1. How did you determine the type of each sentence you created—whether it is declarative, interrogative, imperative, or exclamatory? 2. Among the sentences you and your classmates constructed, which ones can be clearly judged as <i>true</i> or <i>false</i>? Why? 3. How does creating and identifying your own sentence types help you understand the difference between statements and non-statements in logic?	Instructions: 1. Introduce the concept. Briefly review the four sentence types (Declarative, Interrogative, Imperative, Exclamatory). Give one quick example of each type to ensure understanding. 2. Student construction. Instruct students to create their own examples—one for each sentence type—on paper or index cards. Encourage creativity and connection to real-life or mathematical ideas. 3. Sharing and classification. Have students exchange their work with a partner or small group. Each group or partner identifies and labels the sentence type for every example shared. 4. Truth-value analysis. Ask students to determine which of the constructed sentences can be judged as true or false. Use guiding questions such as: ✓ “Can this sentence be proven or disproven?” ✓ “Is it expressing a fact, a command, or a feeling?” 5. Class discussion. Invite volunteers to share examples on the board under each sentence type. As a class, identify which examples can be considered logical propositions. 6. Conclusion. Emphasize that only declarative sentences expressing facts or statements that can be verified as <i>true</i> or <i>false</i> are considered logical propositions in Mathematics. 7. Facilitate reflection through process questions: a. How did you determine the type of each sentence you created—whether it is declarative, interrogative, imperative, or exclamatory? (<i>Encourages learners to recall and apply grammar knowledge as they classify their own constructed sentences.</i>) b. Among the sentences you and your classmates constructed, which ones can be clearly judged as true or false? Why? (<i>Prompts learners to evaluate the truth</i>
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	4. In your own words, how would you describe what a logical proposition is, based on the sentences you created and analyzed?	<i>value of their own examples and distinguish between factual and non-factual statements.)</i> c. How does creating and identifying your own sentence types help you understand the difference between statements and non-statements in logic? <i>(Guides learners to connect their sentence construction activity to the logical distinction between propositions and other sentence forms.)</i> d. In your own words, how would you describe what a logical proposition is, based on the sentences you created and analyzed? <i>(Encourages learners to articulate their personal understanding of logical propositions, reinforcing conceptual comprehension through reflection.)</i> Strategy 2 — Think-Pair-Share (with Sentence Cards) Purpose: To activate learners’ prior knowledge of English sentence types and guide them in constructing their own examples to discover which sentences can be considered logical propositions in Mathematics. Materials: ✓ Whiteboard or projector ✓ writing materials or worksheets ✓ paper or index cards (optional) ✓ timer Instructions: 1. Distribute 3–4 sentence cards to each learner. Each card contains a different sentence type (e.g., declarative, interrogative, imperative, exclamatory). <i>Sample sentences may include:</i> 1. The Philippines is made up of more than 7,000 islands. <i>(Declarative — Proposition: True)</i>
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		2. Is Juan Dela Cruz the head of the curriculum team? (<i>Interrogative — Not a proposition</i>) 3. Please submit your performance task by Friday. (<i>Imperative — Not a proposition</i>) 4. $9 + 4 = 13$. (<i>Declarative — Proposition: True</i>) 5. Everyone loves history class. (<i>Declarative — Not verifiable</i>) 6. Wow, that museum exhibit was incredible! (<i>Exclamatory — Not a proposition</i>) 7. The moon is made of green cheese. (<i>Declarative — Proposition: False</i>) 8. Open your books to page 25. (<i>Imperative — Not a proposition</i>) 9. Chocolate is the best flavor of ice cream. (<i>Declarative — Not verifiable</i>) 10. Cebu is located in the Visayas region. (<i>Declarative — Proposition: True</i>) 2. Explain the task: “You’ll first decide alone, then discuss with a partner, and finally share with the class.” 3. Think: Students work individually. For each sentence, they identify: - Sentence type - Whether it is a proposition (Yes/No) - If Yes, its truth value (True/False) with a short reason 4. Pair: Students compare answers with a partner. Discuss disagreements and reach consensus using reasoning or evidence. Encourage prompts such as: ✓ “How do you know?” ✓ “Can this be checked or verified?” 5. Share: Invite a few pairs to share one sentence where they initially disagreed and explain how they resolved it. Highlight correct reasoning and clarify misconceptions.
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		6. Facilitate reflection through process questions: - How did you determine the type of each sentence discussed in your pair? <i>(Encourages recall and application of grammar knowledge to classification tasks.)</i> - Which sentences were clearly true or false, and how did you know? <i>(Prompts learners to reason through truth values using evidence or logic.)</i> - Why are declarative sentences the only ones suitable as logical propositions in Mathematics? <i>(Leads learners to connect grammatical form with logical reasoning.)</i> - In your own words, how would you describe a logical proposition after completing this activity? <i>(Supports metacognitive reflection and personal articulation of the concept.)</i>
A.2. Establishing the Purpose of the Lesson		
	Activity A.2: Exploring Truth Values of Propositions In this activity, you will explore the idea of truth values in logical propositions. Instead of identifying which sentences can be classified as true or false, you will now be given declarative statements. Your task is to evaluate each one and decide whether it is true or false. This exploration will help you see how propositions work in both real-life situations and in mathematics. Instructions: - You will be given a set of declarative statements. - Water boils at 100 degrees Celsius at sea level. - All Filipino festivals are celebrated in December. - A triangle has four sides.	Activity A.2: Exploring Truth Values of Propositions This activity transitions learners from identifying propositions to evaluating their truth values. The aim is to deepen understanding that propositions are declarative statements that can only be true or false at a given time. Strategy 1: Truth Placard Showdown Purpose: Engage learners actively by using visual response cards (True/False placards). This strategy allows quick checks for understanding and creates an interactive classroom atmosphere. Materials: ✓ Pre-made “True” and “False” placards (or learners can make their own on bond paper) ✓ List of prepared declarative statements Instructions:

	d. Jose Garcia Villa was a Filipino poet and writer. e. The sun rises in the west. f. Bananas grow on trees. g. The capital of Japan is Tokyo. h. Every student in this class owns a smartphone. i. The Pacific Ocean is the smallest ocean in the world. j. Carbon dioxide is a gas used by plants during photosynthesis. 2. Work individually or in pairs to analyze each statement. 3. Decide whether the statement is True or False based on facts or general knowledge. 4. Hold up the “True” or “False” placard (or respond orally) to show your answer. 5. The teacher will guide the discussion, confirming which statements are propositions and clarifying their truth values. Process Questions: - How did you decide whether a statement was True or False? - Why can every proposition only have one truth value at a time? - How does this process help us in reasoning logically in Mathematics?	- Distribute True/False placards to each learner (or pair). - Water boils at 100 degrees Celsius at sea level. <i>True</i> - All Filipino festivals are celebrated in December. <i>False</i> - A triangle has four sides. <i>False</i> - Jose Garcia Villa was a Filipino poet and writer. <i>True</i> - The sun rises in the west. <i>False</i> - Bananas grow on trees. <i>False</i> - The capital of Japan is Tokyo. <i>True</i> - Every student in this class owns a smartphone. <i>False</i> - The Pacific Ocean is the smallest ocean in the world. <i>False</i> - Carbon dioxide is a gas used by plants during photosynthesis. <i>True</i> - Read aloud each declarative statement one by one. - Learners raise the placard showing their evaluation of the statement’s truth value. - Ask a few learners to justify their answers, prompting factual reasoning or logical explanation. - Confirm the correct truth value and provide clarification where misconceptions arise. - Conclude with process questions to encourage reflection and connect to mathematical reasoning. <i>How did you decide whether a statement was True or False?</i> <i>Why can every proposition only have one truth value at a time?</i> <i>How does this process help us in reasoning logically in Mathematics?</i> Strategy 2: Pair-Check-Discuss Purpose: Encourage collaborative reasoning by letting learners test their truth-value judgments against a
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		peer's thinking before sharing with the class. This reduces hesitation and builds confidence. Materials: ✓ Printed list of statements (or written on the board) ✓ Learner notebooks for recording answers Instructions: 1. Ask learners to work individually first, marking each statement as True or False. a. Water boils at 100 degrees Celsius at sea level. <i>True</i> b. All Filipino festivals are celebrated in December. <i>False</i> c. A triangle has four sides. <i>False</i> d. Jose Garcia Villa was a Filipino poet and writer. <i>True</i> e. The sun rises in the west. <i>False</i> f. Bananas grow on trees. <i>False</i> g. The capital of Japan is Tokyo. <i>True</i> h. Every student in this class owns a smartphone. <i>False</i> i. The Pacific Ocean is the smallest ocean in the world. <i>False</i> j. Carbon dioxide is a gas used by plants during photosynthesis. <i>True</i> 2. Afterward, they pair up and compare answers. Where they differ, they must explain their reasoning to each other until reaching consensus. 3. Call on selected pairs to share answers for specific statements and explain their reasoning to the class. 4. Highlight examples where reasoning was strong and clarify incorrect judgments. 5. Wrap up with process questions, emphasizing that each proposition has exactly one truth value. a. How did you decide whether a statement was True or False?
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		b. Why can every proposition only have one truth value at a time? c. How does this process help us in reasoning logically in Mathematics? <i>(These follow-up questions serve as a bridge to the performance task that deepens learners' understanding of logical propositions and truth values in Mathematics.)</i>
B. Instituting New Knowledge	B.1. Presenting Examples	
	Activity B.1: Discovering Patterns in Propositions In this activity, you will explore different kinds of statements. Some stand alone as a single idea, while others combine two or more ideas. By comparing them, you will notice patterns that will help you understand how certain words connect ideas together and form new kinds of statements. Instructions: - Read the following statements carefully: - The Earth revolves around the Sun. - The number 8 is even and the number 9 is odd. - If it snows, then the ground becomes white. - A square has four equal sides. - You will pass the game if and only if you score above 50 points. - The statement is not correct. - A figure is a triangle or it is a quadrilateral. - For each statement: ✓ Decide whether it expresses one idea or more than one idea combined.	Activity B.1: Discovering Patterns in Propositions Strategy 1: Sorting and Highlighting Purpose: Guide learners to distinguish simple from compound propositions by sorting examples and highlighting connecting words, so they discover logical connectives through patterns rather than direct explanation. Materials: ✓ Printed or projected list of statements ✓ highlighters or colored pens (or digital highlighting if using projector/interactive board) ✓ sorting chart with two columns: <i>One Idea / More than One Idea</i> Instructions: - Present the list of statements (a–g) to the class. - Ask learners to work in pairs. First, they classify each statement under One Idea or More than One Idea. - Next, instruct them to underline or highlight any words in the “More than One Idea” group that serve as connectors or modifiers. - Facilitate a class discussion where pairs share examples. List identified connectives on the board (e.g., and, or, if...then, not, if and only if).

- ✓ If it expresses more than one idea, underline or highlight the word(s) that connect or modify the ideas.

Example:

- a. The number 10 is even and the number 7 is odd.

Expresses **more than one idea**.

Connecting word: **and**

→ This is a **compound statement** because it combines two simple statements using *and*.

- b. A triangle has three sides.

Expresses **one idea only**.

No connecting words.

→ This is a **simple statement** because it expresses a single, complete thought.

3. Discuss your answers with your partner and look for patterns.
4. Share your observations with the class.
5. Listen as your teacher explains how statements with one idea differ from those that combine ideas, and how certain connecting words change the meaning of statements.

Process Questions:

1. Which statements seemed to contain only one idea? Which contained more than one?
2. What words served as “clues” that combined or modified the ideas?
3. How did these words change the overall meaning of the statement?

5. Conclude by confirming: Statements with one idea are called simple propositions; those with more than one idea, connected by these words, are compound propositions.
6. Wrap up by asking reflective questions from the activity.
- a. Which statements seemed to contain only one idea? Which contained more than one?
 - b. What words served as “clues” that combined or modified the ideas?
 - c. How did these words change the overall meaning of the statement?
 - d. Why might it be important to know how simple and combined statements work when reasoning logically?

List of Statements:

- a. The Earth revolves around the Sun.
One Idea
- b. The number 8 is even **and** the number 9 is odd.
More than One Idea
- c. **If** it snows, **then** the ground becomes white.
More than One Idea
- d. A square has four equal sides.
One Idea
- e. You will pass the game **if and only if** you score above 50 points.
More than One Idea
- f. The statement is **not** correct.
More than One Idea
- g. A figure is a triangle **or** it is a quadrilateral.
More than One Idea

Strategy 2: Think-Pair-Compare

Purpose: Encourage learners to analyze statements individually, refine their reasoning through peer collaboration, and then compare results with the class, promoting deeper understanding of logical connectives.

4. Why might it be important to know how simple and combined statements work when reasoning logically?

Materials:

- ✓ Index cards (one statement per card)
- ✓ “Clue word” chart posted on the board (blank at first, later filled with learners’ discoveries)

Instructions:

1. Distribute one statement card to each learner. Ask them to decide individually if their statement is *one idea* or *more than one idea combined*.
2. Pair them with a partner to compare answers and justify their reasoning. If they disagree, they must explain their thinking until they reach consensus.
3. Invite pairs to come forward and place their card under *Simple* or *Compound* on the board.
4. As learners present, highlight the connective words that appear in compound propositions, adding them to the “Clue word” chart.
5. Synthesize findings: guide the class to see how these connectives form compound propositions, then formally introduce the concept of logical operations.
6. End with reflection questions, connecting the discoveries to mathematical reasoning.
 - a. Which statements seemed to contain only one idea? Which contained more than one?
 - b. What words served as “clues” that combined or modified the ideas?
 - c. How did these words change the overall meaning of the statement?
 - d. Why might it be important to know how simple and combined statements work when reasoning logically?

List of Statements:

- a. The Earth revolves around the Sun.
Simple
- b. The number 8 is even **and** the number 9 is odd.

		<i>Compound</i> c. If it snows, then the ground becomes white. <i>Compound</i> d. A square has four equal sides. <i>Simple</i> e. You will pass the game if and only if you score above 50 points. <i>Compound</i> f. The statement is not correct. <i>Compound</i> g. A figure is a triangle or it is a quadrilateral. <i>Compound</i>
	B.2. Discussing the Concept	
	Activity B.2: Deepening Understanding on Propositions Instructions: To be able to understand the key concepts of the lesson, listen attentively at participate actively to the discussion. Propositions A logical statement, also called a proposition, is a declarative sentence that is either true or false, but never both. If the proposition is correct, it has a truth value of “true”; otherwise, its truth value is “false.” In symbolic logic, a proposition is usually represented by a small letter. For example: Let a denote the proposition: “Noli Me Tangere is written by Dr. Jose P. Rizal.” This can be read as: a is the proposition “Noli Me Tangere is written by Dr. Jose P. Rizal.”	Activity B.2: Deepening Understanding on Propositions Strategy 1: Pattern Discovery Strategy Purpose: To help learners discover the difference between simple and compound propositions by observing and analyzing examples. How it works: 1. Present a mixed set of sentences on the board (some simple, some compound). 2. Ask learners to classify them into groups based on observed patterns, without giving the definition first. 3. Guide them through questions that lead to the generalization of the concepts. Example: 1. Statements: 1. "The sky is blue."

	When dealing with multiple propositions, we may use subscripts, such as p_1, p_2, p_3, ..., to denote a sequence of propositions. Example: Examine each sentence carefully. Decide whether it is a proposition or not. If it is a proposition, state whether its truth value is true or false. a: The sun rises in the east. → Proposition (True) b: What time is it now? → Not a proposition (a question) c: $2 + 3 = 6$. → Proposition (False) d: Close the door. → Not a proposition (a command) e: Manila is the capital city of the Philippines. → Proposition (True) p: All triangles have three sides. → Proposition (True) q: Please pass the paper. → Not a proposition (command) p_1: The Earth has two moons. → Proposition (False) p_2: Is it raining outside? → Not a proposition (question) p_3: 7 is a prime number. → Proposition (True)	2. "The ground is wet and the sky is cloudy." 3. "If it rains, then the ground will be wet." 2. Guiding Questions: 1. Which statements contain only one idea? 2. Which statements combine two or more ideas? 3. What words or symbols show the combination? Discussion Wrap-up: ○ Learners generalize that: 1. Simple propositions contain a single fact or idea. 2. Compound propositions join two or more simple propositions with connectives such as <i>not</i>, <i>and</i>, <i>or</i>, <i>if-then</i>, and <i>if and only if</i>. ○ Teacher confirms and formalizes the definition. Mastery: Students realize that logical connectives are the “clues” that turn a simple proposition into a compound one. Strategy 2: Graphic Organizer Strategy Purpose: To provide a clear and visual distinction between simple and compound propositions. How it works: 1. Prepare a two-column chart on the board or projector: Simple Propositions vs. Compound Propositions.
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	Simple and Compound Propositions In logic, propositions are statements that can be clearly identified as either <i>true</i> or <i>false</i>. They form the building blocks of logical reasoning. Simple Proposition A Simple Proposition is a single, indivisible statement. It expresses one idea and cannot be broken down into smaller propositions. Example: • p: "The sky is blue." • This is a simple proposition because it stands alone and conveys a complete thought. Compound Proposition A Compound Proposition is formed by combining two or more simple propositions using logical connectors (also called logical operators). Example: • p: "The sky is blue." • q: "It is daytime." • p and q: "The sky is blue and it is daytime." • This is a compound proposition because it joins two ideas. Example: Distinguishing Simple and Compound Propositions Simple Propositions (cannot be broken into smaller statements): 1. The sky is blue. 2. Jose Rizal was born in Calamba, Laguna. 3. Water boils at 100°C. 4. Manila is the capital of the Philippines.	2. As a class, sort examples into the correct category. 3. Introduce logical symbols (p, q, $\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$) alongside the sentences. Example: 1. Simple: ■ "It is sunny." (p) ■ "Manila is the capital of the Philippines." (q) 2. Compound: ■ "It is sunny or it is raining." ($p \vee q$) ■ "If it rains, then the ground will be wet." ($p \rightarrow q$) Discussion Wrap-up: 3. Show that compound propositions always depend on connectives, while simple propositions stand on their own. 4. Use colors or arrows to highlight the role of each logical operator. Mastery: Learners can now easily classify statements, symbolize them, and see the structure of compound propositions. Strategy 3: Interactive Truth Table Strategy Purpose: To demonstrate that truth tables are tools for analyzing compound propositions, but not needed for simple ones. How it works: 1. Start with a simple proposition: ■ "p: The number is even." $\rightarrow$ Truth value is either T or F (no table needed). 2. Move to a compound proposition:
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5. Two is an even number.

Compound Propositions (formed by combining two or more simple propositions):

1. The sun is shining and the birds are singing.
2. Either the students are inside the classroom or they are at the library.
3. It is not true that Baguio is the capital of the Philippines.
4. If it rains, then the ground will be wet.
5. You will pass the exam if and only if you study well.

Logical Operators

Logical operators are essential components of propositional logic used to combine individual statements into compound propositions. They express relationships between the **truth values** (true or false) of statements, allowing us to analyze how different conditions interact logically. By applying these operators, we can construct more complex arguments and evaluate their validity.

The **fundamental logical operators** include:

A. Negation

Negation is a logical operation that reverses the truth value of a given statement. The negation of a statement **p** is written as **~p**, where the symbol **~** means “**not**.”

Property of Negation

- If **p** is **true**, then **~p** is **false**.
- If **p** is **false**, then **~p** is **true**.

p	~p
T	F
F	T

- " $p \wedge q$: The number is even and divisible by 3."
3. Construct the truth table step by step with the class.

Discussion Wrap-up:

- Emphasize that truth tables are essential for **compound propositions** because they involve logical operators.
- Reinforce those simple propositions **don't require truth tables** since they have a direct truth value.

Mastery:

Students gain confidence in analyzing compound statements systematically using truth tables.

Example 1: State the negation of each of the following propositions.

- p_1 : Proper waste segregation reduces pollution.
- p_2 : Reforestation does not restore biodiversity.
- p_3 : Clean energy does not reduce carbon emissions.
- p_4 : Environmental education promotes sustainable practices.
- p_5 : Community action prevents ecological degradation.

Solution:

- $\sim p_1$: Proper waste segregation **does not** reduce pollution.
- $\sim p_2$: Reforestation restores biodiversity.
- $\sim p_3$: Clean energy reduces carbon emissions.
- $\sim p_4$: Environmental education does not promote sustainable practices.
- $\sim p_5$: Community action does not prevent ecological degradation.

B. Conjunction

Conjunction is a logical operation that combines two statements, p and q , into a single compound statement: “**p and q**”, symbolically written as $p \wedge q$, where $\wedge$ denotes the logical operator **AND**.

This operator asserts that **both conditions must be true** for the compound statement to be true.

Property of Conjunction

- The compound statement $p \wedge q$ is **true only if** both **p** and **q** are true.
- If either **p** or **q** is false—or both are false—then $p \wedge q$ is **false**.

p	q	$p \wedge q$
T	T	T

T	F	F
F	T	F
F	F	F

Example 2: Let the propositions be:

p: Mental health support fosters resilience.

q: Lack of awareness leads to stigma.

Express the following using verbal sentences and symbolic logic:

1. $p \wedge q$
2. $p \wedge \sim q$
3. Mental health support does not foster resilience and lack of awareness does not lead to stigma.
4. Mental health support does not foster resilience and lack of awareness leads to stigma.

Solution:

1. Mental health support fosters resilience and lack of awareness leads to stigma.
2. Mental health support fosters resilience and it is not true that lack of awareness leads to stigma.
3. $\sim p \wedge \sim q$
4. $\sim p \wedge q$

C. Disjunction

Disjunction is a logical operation that connects two statements, p and q , into a compound statement: “**p or q**”, symbolically written as $p \vee q$, where $\vee$ represents the logical operator **OR**.

This operator expresses that **at least one** of the statements must be true for the compound statement to be true.

Property of Disjunction

- The statement $p \vee q$ is **true** if:
 - p is true,
 - q is true,
 - or **both** are true.
- It is **false only** when **both p and q are false**.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Example 3: Let the propositions be:

a: Everyone deserves access to quality education.

b: Marginalized voices must be heard.

c: Systemic inequality must be addressed.

d: Community action promotes equity.

e: Policy reform ensures justice.

Express the disjunctions below in verbal sentences or in symbols.

1. $a \vee b$
2. $(\sim a \wedge \sim b) \vee c$
3. $\sim d \wedge (\sim c \vee b)$
4. Either everyone deserves access to quality education, or marginalized voices must be heard and systemic inequality must be addressed.
5. Either everyone does not deserve access to quality education or policy reform does not ensure justice, and community action promotes equity.

Solution:

1. $a \vee b$

Verbal: Either everyone deserves access to quality education **or** marginalized voices must be heard.

2. $(\sim a \wedge \sim b) \vee c$

Verbal: Everyone **does not** deserve access to quality education **and** marginalized voices **are not** heard, **or** systemic inequality must be addressed.

3. $\sim d \wedge (\sim c \vee b)$

Verbal: Community action does not promote equity, and either systemic inequality is not addressed or marginalized voices must be heard.

4. $a \vee (b \wedge c)$

5. $(\sim a \vee \sim e) \wedge d$

D. Conditional (Implication)

A **conditional statement**, also called an **implication**, connects two propositions, **p** and **q**, into a compound statement: “**If p, then q**”, symbolically written as $p \rightarrow q$, where $\rightarrow$ represents the logical operator **IF-THEN**.

- **p** is called the **hypothesis** or **premise**.
- **q** is called the **conclusion** or **consequence**.

This structure expresses that **if the first condition (p) holds**, then the second condition (q) must follow.

Property of Conditional Statements

The conditional statement $p \rightarrow q$ is:

- **False only** when **p is true** and **q is false**.
- **True** in all other cases.

p	q	$p \rightarrow q$
T	T	T
T	F	F

F	T	T
F	F	T

Example 4: Let the propositions be:

x: A number is divisible by 2.

y: The number is divisible by 4.

z: The number is even.

Express the conditionals below in verbal sentences or symbols, as the case may be:

1. $x \rightarrow z$
2. $\sim y \rightarrow (\sim x \wedge \sim z)$
3. $z \rightarrow (x \wedge y)$
4. If a number is not divisible by 2, then it is not true that the number is divisible by 4 or that it is even.
5. If a number is divisible by 2 but not divisible by 4, then the number is not even.

Solution:

1. $x \rightarrow z$
Verbal: **If** a number is divisible by 2, **then** the number is even.
2. $y \rightarrow (\sim x \wedge \sim z)$
Verbal: If a number is not divisible by 4, then it is not divisible by 2 and it is not even.
3. $z \rightarrow (x \wedge y)$
Verbal: **If** a number is even, **then** it is divisible by 2 **and** divisible by 4.
4. $\sim x \rightarrow \sim(y \vee z)$
5. $(x \wedge \sim y) \rightarrow \sim z$

E. Biconditional

A **biconditional statement** connects two propositions, **p** and **q**, into a compound statement: “**p if and only if q**”, symbolically written as **$p \leftrightarrow q$** , where $\leftrightarrow$ represents the logical operator **IF AND ONLY IF**.

This means that **p and q must always share the same truth value**—either both true or both false—for the biconditional to be true.

- **p** is true **if and only if q** is true.
- The relationship is **mutual** and **equivalent**.

Property of Biconditional Statements

- **$p \leftrightarrow q$** is **true** when **both p and q are true**, or **both are false**.
- It is **false** when **p and q have opposite truth values**.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Example 5: Let the propositions be:

p: A polygon has more than three sides.

q: The sum of its interior angles exceeds 180 degrees.

r: The polygon is not a triangle.

Express the biconditionals below in verbal sentences or symbols, as the case may be:

1. **$p \leftrightarrow r$**
2. **$\sim q \leftrightarrow (\sim p \wedge \sim r)$**
3. **$r \leftrightarrow (p \wedge q)$**

4. A polygon does **not** have more than three sides **if and only if** it is **not true** that the sum of its interior angles exceeds 180 degrees **or** that it is not a triangle.
5. A polygon has more than three sides **but** its interior angles do **not** exceed 180 degrees **if and only if** it is a triangle.

Solution:

1. $p \leftrightarrow r$
Verbal: A polygon has more than three sides **if and only if** it is not a triangle.
2. $\sim q \leftrightarrow (\sim p \wedge \sim r)$
Verbal: The sum of a polygon's interior angles does **not** exceed 180 degrees **if and only if** it does **not** have more than three sides **and** it **is** a triangle.
3. $r \leftrightarrow (p \wedge q)$
Verbal: A polygon is not a triangle **if and only if** it has more than three sides **and** the sum of its interior angles exceeds 180 degrees.
4. $\sim p \leftrightarrow \sim(q \vee r)$
5. $(p \wedge \sim q) \leftrightarrow \sim r$

Constructing Truth Tables for Logical Operators

A **truth table** is a mathematical table used in logic to determine the truth value of a compound proposition by listing all possible truth values of its components.

It helps us see how **logical operators** (AND, OR, NOT, etc.) affect propositions.

Example 6: 1. $(p \rightarrow q) \wedge (p \rightarrow \sim q)$

Step-by-step:

- Compute $p \rightarrow q$ (false only if $p = T, q = F$).
- Compute $\sim q$ (negation of q).
- Compute $p \rightarrow \sim q$.
- Finally, take the AND of $(p \rightarrow q)$ and $(p \rightarrow \sim q)$

p	q	$p \rightarrow q$	$\sim q$	$p \rightarrow \sim q$	$(p \rightarrow q) \wedge (p \rightarrow \sim q)$
T	T	T	F	F	F
T	F	F	T	T	F
F	T	T	F	T	T
F	F	T	T	T	T

Example 7: $(p \rightarrow q) \leftrightarrow (\sim p \vee q)$

Step-by-step:

- Compute $p \rightarrow q$.
- Compute $\sim p$.
- Compute $\sim p \vee q$.
- Biconditional $\leftrightarrow$ is true when both sides are the same.

p	Q	$p \rightarrow q$	$\sim p$	$\sim p \vee q$	$(p \rightarrow q) \leftrightarrow (\sim p \vee q)$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	T
F	F	T	T	T	T

B.3. Developing Mastery**Activity B.3: Building Blocks of Logic**

In this activity, you will practice working with propositions and logical operators. First, you will identify whether a statement is a proposition and decide if it is true or false. Then, you will classify statements as either

Activity B.3: Building Blocks of Logic

This activity is designed to help learners practice and master the concepts of propositions, simple and compound statements, logical operators, and truth tables. It provides a progression from recognition of

	simple or compound and spot the logical operators used. Finally, you will apply what you learned by writing propositions in symbols and creating truth tables. This activity will help you master how to analyze and combine statements in logic step by step. I. Spot the Proposition Directions: For each statement, decide whether it is a proposition or not. If it is a proposition, state its truth value (True or False). 1. The Earth revolves around the Sun. 2. What time is your class tomorrow? 3. Close the window. 4. $2 + 5 = 8$. 5. Manila is the capital of the Philippines. 6. Do your homework. <i>Guide Questions:</i> 1. Is it a declarative sentence? 2. Can it be clearly labeled as true or false? II. Sorting Simple and Compound Directions: Classify the following as Simple or Compound Propositions. If compound, identify the type of connective(s) used. 1. The sky is clear. 2. The sky is clear and the air is fresh. 3. If it rains, then the ground will be wet. 4. Either the lights are off or the power is interrupted. 5. Jose Rizal is a national hero. 6. The school is open if and only if it is Monday. III. Operator Hunt	propositions to symbolic representation and logical analysis. Strategy 1: Independent Practice (Seatwork Drill) Purpose: Strengthen individual accountability by allowing learners to recognize, classify, and analyze propositions independently, ensuring personal mastery of logical concepts. Materials: ✓ Activity sheets (with Parts A–D: propositions, simple/compound, operators, truth tables) ✓ Pen/pencil Instructions: 1. Distribute activity sheets to each learner. 2. Give them time to answer independently, focusing first on identifying propositions, then progressing to distinguishing types and applying operators. 3. Remind learners that they should not look at others' work during this round, to assess their personal understanding. 4. Collect the sheets or check together after completion, inviting volunteers to write answers on the board. 5. Facilitate class discussion by clarifying misconceptions and emphasizing correct reasoning. Reflection Questions: a. Which items were easiest for you to recognize? Why? <i>(Encourages learners to reflect on which statements they understood quickly and what helped their comprehension.)</i> b. Which items were most confusing? What made them difficult? <i>(Prompts learners to identify areas</i>
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	Directions: Identify the logical operator(s) used in each proposition below and rewrite them using symbols. 1. It is not true that the number is even. 2. The students are noisy and the teacher is absent. 3. If the bell rings, then the class will be dismissed. 4. You will pass if and only if you study. 5. Either the exam is today or it is tomorrow. <i>Expected Output Example:</i> • Sentence: “It is not true that the number is even.” • Operator: Negation ($\sim$) • Symbolic Form: $\sim p$ IV. Truth Table Challenge Directions: Construct the truth tables for the following compound propositions: 1. $\sim p \vee q$ 2. $(p \wedge q) \rightarrow \sim p$ 3. $(p \vee q) \leftrightarrow (\sim p \wedge q)$ <i>Guide Questions:</i> a. How many rows will the table have for 2 variables? b. What is the truth value of the compound statement in each case? Process Questions: 1. How did this activity help you understand what makes a statement logical or not? 2. What clues helped you decide whether a statement is true or false?	<i>of misunderstanding and the reasons behind their confusion.)</i> c. How did you decide whether a statement was simple or compound? <i>(Guides learners to articulate their reasoning process in identifying and classifying statements.)</i> d. How confident are you in constructing truth tables on your own? <i>(Allows learners to self-assess their readiness and confidence in applying the concept independently.)</i> Strategy 2: Think–Pair–Compare Purpose: Encourage learners to analyze statements individually, refine their reasoning through peer collaboration, and then compare results with the class, promoting deeper understanding of logical connectives. Materials: ✓ Index cards (one statement per card) ✓ “Clue word” chart posted on the board (blank at first, later filled with learners’ discoveries) Instructions: 1. Distribute one statement card to each learner. Ask them to decide individually if their statement is one idea or more than one idea combined. 2. Pair them with a partner to compare answers and justify their reasoning. If they disagree, they must explain their thinking until they reach consensus. 3. Invite pairs to come forward and place their card under Simple or Compound on the board. 4. As learners present, highlight the connective words that appear in compound propositions, adding them to the “Clue word” chart.
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	3. How did you determine the relationships or connections between different propositions? 4. Why are logical operators important in forming clear and meaningful statements? 5. How can using logic help you make better decisions or avoid false information in real life?	5. Synthesize findings: guide the class to see how these connectives form compound propositions, then formally introduce the concept of logical operations. 6. End with reflection questions, connecting the discoveries to mathematical reasoning. Reflection Questions: a. Which statements seemed to contain only one idea? Which contained more than one? <i>(Encourages learners to distinguish between simple and compound statements through observation and comparison.)</i> b. What words served as “clues” that combined or modified the ideas? <i>(Helps learners identify logical connectives or linking words as key indicators of compound statements.)</i> c. How did these words change the overall meaning of the statement? <i>(Leads learners to understand how connectives affect the relationship and truth conditions between ideas.)</i> d. Why might it be important to know how simple and combined statements work when reasoning logically? <i>(Prompts learners to connect the concept of statement structure to logical reasoning and problem-solving in Mathematics.)</i> Strategy 3: Group Challenge (Collaborative Gamified Approach) Purpose: Build teamwork and engagement by turning logical analysis into a group challenge, allowing learners to practice applying logical operators and constructing truth tables in a supportive, competitive setting. Materials: ✓ Manila paper or whiteboard space for each group ✓ Markers
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		✓ Pre-assigned sets of propositions (different per group) Instructions: 1. Divide the class into small groups (3–5 learners). 2. Assign each group a different set of statements. Ask them to: a. Classify each as proposition or not. b. Sort simple vs. compound. c. Identify the logical operator(s). d. Construct truth tables for the compound ones. 3. Give groups 10–15 minutes to solve collaboratively on manila paper/board. 4. Groups post their outputs on the wall/board and present briefly. 5. Award points for accuracy, clarity, and teamwork. Reflection Questions: a. How did working in a group help you solve the problems? <i>(Encourages learners to reflect on collaboration, communication, and shared reasoning in problem-solving.)</i> b. Which part of the activity did your group find most challenging? <i>(Helps identify learning difficulties and guide future scaffolding or clarification.)</i> c. How did the logical operators change the truth values of the statements? <i>(Leads learners to analyze the effect of connectives such as AND, OR, and NOT on compound propositions.)</i> d. What strategies did your group use to complete the truth tables? <i>(Promotes metacognitive reflection on systematic and cooperative approaches to constructing truth tables.)</i>
C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Logic in Action Stand Up with Reason	Activity C.1: Logic in Action Stand Up with Reason

	In this activity, you will perform a short role play showing how logical thinking can help resolve real-life issues involving bullying or mental health. This is to apply the principles of logical propositions and connections to evaluate arguments and promote respectful, reasoned communication. Instructions: 1. Form a group of 3–5 members. 2. Prepare a 2–3 minute role play about a real-life situation involving bullying or mental health (e.g., online teasing, peer conflict, or misunderstanding). 3. Include a part where someone uses faulty reasoning, then show how logic and reasoning help clarify the situation. 4. Use simple logical connectors such as <i>if...then...</i>, <i>either...or...</i>, or <i>not both...</i>. 5. End your skit with a positive and respectful resolution that supports mental wellness. Process Questions: 1. What wrong reasoning or assumption did you show in your role play? 2. How was logical thinking used to fix the misunderstanding? 3. How can logical communication help prevent bullying or protect mental health? Criteria for Scoring: <table><tr><th>Criteria</th><th>Points</th></tr><tr><td>Use of logical propositions and reasoning</td><td>10</td></tr><tr><td>Relevance to mental health and bullying</td><td>5</td></tr><tr><td>Clarity and organization of message</td><td>5</td></tr></table>	Criteria	Points	Use of logical propositions and reasoning	10	Relevance to mental health and bullying	5	Clarity and organization of message	5	Purpose: This activity allows learners to apply logical propositions and connections in real-life contexts by dramatizing common issues related to bullying and mental health. Through role play, students can demonstrate how logic helps clarify misunderstandings, resolve conflicts, and promote empathy and emotional well-being. Materials: ✓ Prepared role play scenario (optional) ✓ Scoring sheet ✓ Timer or stopwatch ✓ Simple props (optional, for engagement) Instructions: 1. Introduce the activity by connecting it to the lesson’s objective: <i>evaluating arguments using principles of logical propositions and logical connections in real-life contexts</i>. 2. Discuss how logical thinking can help resolve misunderstandings or harmful assumptions, especially in situations involving bullying or mental health. 3. Present Activity C.1: Logic in Action – Stand Up with Reason and review the learner instructions. 4. Divide the class into small groups of 3–5 members. Give them 10–15 minutes to plan and rehearse their 2–3 minute role play. 5. Remind them to include one instance of faulty reasoning and show how logic can correct it using connectors like <i>if...then...</i> or <i>either...or...</i> 6. Facilitate presentations and ensure that each group stays within the time limit. 7. After each performance, guide the class in answering the process questions and emphasize
Criteria	Points									
Use of logical propositions and reasoning	10									
Relevance to mental health and bullying	5									
Clarity and organization of message	5									

	Teamwork and participation	5	the importance of combining logic with empathy and respect.
	Creativity and engagement	5	
	TOTAL	30	
			8. Use the provided rubric to assess performance based on logic application, relevance, clarity, teamwork, and creativity.
			9. Facilitate the process questions by guiding students in a short reflection discussion.
			a. What wrong reasoning or assumption did you show in your role play? <i>(Encourages learners to identify and analyze the type of faulty reasoning or fallacy presented in their scenario.)</i>
			b. How was logical thinking used to fix the misunderstanding? <i>(Prompts learners to explain how applying logical propositions and connections helped clarify the issue or resolve the conflict.)</i>
			c. How can logical communication help prevent bullying or protect mental health? <i>(Leads learners to reflect on the importance of logic, empathy, and respectful communication in maintaining positive relationships and emotional well-being.)</i>
			Criteria for Scoring:
			Criteria
			Points
			Use of logical propositions and reasoning
			10
			Relevance to mental health and bullying
			5
			Clarity and organization of message
			5
			Teamwork and participation
			5
			Creativity and engagement
			5
			TOTAL
			30
C.2. Making Generalization			
Activity C.2: Logic Web of Understanding		Activity C.2: Logic Web of Understanding	
		This activity allows learners to consolidate their	

	In this activity, you will summarize and connect what you have learned about propositions and logical operators. Working with your group, you will create a Logic Web that shows how simple and compound propositions are related, how logical operators combine ideas, and how truth tables help test the validity of compound statements. By the end, you will use your Logic Web to make a general statement about the importance of logic in reasoning and decision-making. Instructions: 1. Recall – Review the concepts: propositions, simple vs. compound, and the five logical operators. 2. Group Work – On a manila paper/cartolina, draw a web/diagram starting with “Propositions” at the center. - o Branch 1: Define and give examples of Simple Propositions. - o Branch 2: Define and give examples of Compound Propositions. - o Branch 3: Show the Logical Operators ($\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) with at least one example each. - o Branch 4: Demonstrate how a truth table works using one compound proposition. 3. Synthesize – At the bottom of your web, write one generalization that captures the essence of logical reasoning (e.g., “<i>Logic provides a structured way of making decisions by combining statements and analyzing their truth values.</i>”). 4. Share – Present your group’s Logic Web and generalization to the class. Guide Questions for Generalization a. What is the difference between a simple and compound proposition?	understanding of logical propositions, logical operators, and truth tables. It promotes synthesis by requiring them to visually map connections and formulate a generalization about the role of logic in reasoning and decision-making. Preparation: 1. Prepare manila paper/cartolina and markers (one set per group). 2. Review key concepts (propositions, operators, truth tables) before the activity to ensure learners have the necessary foundation. 3. Display a sample “mini web” to guide learners, but avoid giving the full answer to encourage discovery. Implementation Steps: 1. Recall Phase: Briefly review propositions and operators with the class to activate prior knowledge. 2. Group Work: Divide learners into small groups (4–6 members). Distribute materials and instruct them to construct their Logic Web following the given branches. 3. Facilitation: Move around the classroom to monitor group discussions, provide guiding questions, and clarify misconceptions. Encourage creativity in how they illustrate connections. 4. Synthesis: Remind groups to write a clear generalization at the bottom of their web. Highlight that this should connect logic to reasoning in real-life situations. 5. Sharing: Have each group present their web briefly. Encourage other groups to ask clarifying questions or add insights. Assessment/Reflection:
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	b. How do logical operators change the meaning of propositions? c. Why are truth tables important in analyzing compound statements? d. How does logic help us reason clearly and avoid faulty conclusions?	• Use the Guide Questions for Generalization to facilitate a class discussion after presentations. • Check if learners' outputs show accurate definitions, correct use of logical operators, and at least one correctly constructed truth table. • Provide feedback emphasizing both content accuracy and clarity of presentation.
C.3. Evaluating Learning		
	Activity C.3: Assessment on Logical Propositions In this activity, you will show what you have learned about logical propositions and operators. You will evaluate given statements, decide if they are propositions or not, classify them as simple or compound, and translate them into symbols. You will also practice combining two or more statements using logical connectors such as and ($\wedge$), or ($\vee$), not ($\sim$), if...then ($\rightarrow$), and if and only if ($\leftrightarrow$). Finally, you will construct truth tables to test whether compound statements are true or false. This will help you check how well you understand logic and how to apply it to reasoning clearly. Part I. True or False. Determine if each statement is True or False. 1. Baguio is the capital of the Philippines. 2. 10 is an even number. 3. All squares are rectangles. 4. The moon is made of cheese. 5. If it rains, then the ground will be wet.	Activity C.3: Assessment on Logical Propositions This activity is designed to check learners' understanding of logical propositions and operators. It combines recognition, classification, symbolization, and truth table construction. The teacher should guide the learners step by step but allow them to apply reasoning independently. Strategy 1: Individual Written Assessment 1. Distribute the assessment sheets to each learner. 2. Give clear time allotments for each part (e.g., 5 minutes for Parts I & II, 10 minutes for Parts III & IV, and 15 minutes for Part V). 3. Remind learners to show their reasoning where necessary, especially for truth table construction. 4. Collect the papers for checking or discuss answers afterward for immediate feedback.

Part II. Proposition or Not. State whether each is a Proposition or Not a Proposition. If it is a proposition, determine its truth value (True or False).

6. What is your favorite subject?
7. The Earth has one moon.
8. Please close the window.
9. $5 + 7 = 13$.
10. The Pacific Ocean is larger than the Atlantic Ocean.

Part III. Simple or Compound. Classify each as a Simple Proposition or Compound Proposition.

11. A triangle has three sides.
12. It is raining and the wind is strong.
13. If you study well, then you will pass the exam.
14. The classroom is clean.
15. Jose Rizal is the national hero or Andres Bonifacio is the national hero.

Part IV. Using Logical Operators

Express the following in **symbols**. Let:

p: The sky is blue.

q: It is daytime.

r: The sun is shining.

16. If it is daytime, then the sun is shining.
17. The sky is not blue.
18. The sun is shining or it is daytime.
19. The sky is blue if and only if it is daytime.
20. $(p \leftrightarrow q) \wedge \sim r$

Part V. Truth Table. Construct the truth table for the compound proposition.

21. $(p \wedge \sim q) \rightarrow (\sim p \leftrightarrow q)$
- 22.

Process Questions:

	1. What steps did you follow to decide whether a statement is true or false and whether it is a proposition? 2. How can you tell a declarative sentence (proposition) apart from a question or a command? Give one example from the worksheet. 3. How did combining simple statements with words like and, or, if...then, or if and only if change the meaning of the original statements? 4. How did writing sentences with symbols (p, q, r, $\sim$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) help you understand their logical relationships? 5. What did the truth table show you about the overall truth of the compound statement — when is a compound always true, always false, or sometimes true?	
	C.4. Additional Activities Activity C.4: Reinforcement on Logical Propositions In these activities, you will get the chance to practice more and go deeper in understanding logical propositions. These additional activities will allow you to review what you find difficult and at the same time explore more advanced applications of logic, making your learning more complete and practical. Remediation Exercises I. Proposition or Not? Directions: Identify whether each statement is a proposition or not. If it is a proposition, write its truth value (True or False). 1. The sun rises in the east. 2. Is the Earth flat? 3. Mathematics is an easy subject. 4. Please close the door. 5. Ten is greater than five. Activity C.4: Reinforcement on Logical Propositions The following activities are designed to address diverse learner needs after the evaluation stage. They serve two purposes: Remediation for learners who need more practice in mastering the basics, and Enhancement for learners who are ready to apply their understanding of logical propositions in more complex and real-life contexts. Remediation Exercises These tasks focus on reinforcing fundamental skills such as identifying propositions, distinguishing between simple and compound statements, and correctly using logical operators. Teachers should: 1. Guide learners step by step, especially in determining whether a statement is a proposition. 2. Provide concrete examples or re-teach the concept before giving the exercises if necessary.	

	II. Simple vs. Compound Sorting Directions: Write S if the statement is <i>simple</i> and C if it is <i>compound</i>. 1. The Philippines is an archipelago. 2. Cebu is a province and Manila is a city. 3. The classroom is clean. 4. Jose Rizal is a national hero or Andres Bonifacio is the Father of the Philippine Revolution. 5. A triangle has three sides. III. Fill-in-the-Operator Directions: Combine the given propositions using the indicated operator. 1. p: "It is raining." q: "I will bring an umbrella." - o Combine using $\wedge$ (and). 2. p: "She studies hard." q: "She will pass the test." - o Combine using $\rightarrow$ (if...then). 3. p: "The school canteen is open." q: "The students will not buy outside." - o Combine using $\leftrightarrow$ (if and only if). Enhancement Exercises I. Real-Life Logic Analysis Directions: Rewrite the following into symbolic form using p, q, r as needed. 1. If a student studies hard, then they will pass. 2. A school is safe if and only if it has security guards. 3. The library is open or the laboratory is available. II. Truth Table Challenge Directions: Construct a truth table for the given compound propositions. 1. $(p \wedge q) \rightarrow r$ 2. $(p \vee q) \wedge \sim p$	3. Check answers together as a class to clarify misconceptions. Enhancement Exercises These tasks extend learning by allowing students to apply logic in symbolic form, construct truth tables, and analyze real-life arguments. Teachers should: 1. Encourage learners to think critically and independently. 2. Allow group or pair work for the "Debate with Logic" activity to foster collaboration and deeper discussion. 3. Challenge learners to explain not just the answers but also their reasoning process. Assessment Strategy ✓ Use the Remediation activities as short drills or seatwork for struggling learners. ✓ Use the Enhancement activities as enrichment tasks for advanced learners or as extension activities for the whole class. ✓ Provide constructive feedback, highlighting both correct reasoning and areas for improvement.
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	III. Debate with Logic Directions: Express each argument using propositions and logical operators. 1. If students are disciplined, then the classroom is peaceful. 2. If teachers are absent or students are noisy, then learning is disrupted. 3. A learner is active if and only if he participates in class discussions.	
III. CONTENT	Lesson 12.2. Conditional Propositions Recommended Time Allotment: 2 Hours	
IV. OBJECTIVES	At the end of the lesson, the learners... 1. define the forms of conditional propositions; 2. identify the form of conditional proposition used in a given statement; 3. construct the converse, inverse, and contrapositive of a given conditional statement using sentences and symbols; and 4. determine the truth values of conditional statements and their related forms in real-world contexts (e.g., laws, classroom rules, common statements).	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ANNOTATION (Instructions for Teachers)
A. Activating Prior Knowledge	A.1. Eliciting Prior Knowledge	
	Activity A.1: Negate the Statement Instructions: 1. You will be given strips of paper containing three statements. List of statements to be used: • Dogs are mammals. • 5 is an even number. • Today is a sunny day. • Jose Rizal was born in 1861. • The sun is white. • Triangle is a four-sided polygon. • A senator is a lawmaker.	Activity A.1: Negate the Statement This activity reviews learners’ understanding of negation, a key logical operation in constructing the converse, inverse, and contrapositive of conditional propositions. Through this activity, you will be able to assess your learners’ readiness on the lesson. Prepare a list of statements that you can use for this activity. You may consider the statements on the left. Note: The teacher may use other propositions of his/ her own.

	2. Your task is to rewrite each statement in its negated form. 3. Be sure that the meaning is correctly flipped, and the negation applies to the whole statement. 4. You will be given three minutes to complete the task. 5. Work carefully and be ready to share your answers afterward. Processing Questions: 1. What word/s do you use to negate a statement? 2. When the statement is true, is it still true when you negate it? Explain your answer. 3. When the statement is false, is it still false when you negate it? Explain your answer.	To facilitate this, you may use any of these two strategies that best fit your classroom context: Strategy 1: Think-Pair-Share (for small classes) 1. Give each learner a strip of paper with three statements on it. 2. Let them think first individually about how to negate the statements. 3. Have them find a pair to compare and to refine their answers. 4. A few pairs will share their work in the class for discussion. Strategy 2: Group Work (for larger classes) 1. Divide the class into small groups of 4–6 members. 2. Each group will receive a strip of paper containing three statements. 3. Group members will collaborate to negate the statements. 4. Each group will choose one representative to present their answers to the class. Answer: Dogs are not mammals. 5 is not an even number. Today is not a sunny day. Jose Rizal was not born in 1861. The sun is not white. Triangle is not a four-sided polygon. A senator is not a lawmaker.
	A.2. Establishing the Purpose of the Lesson	
	Activity A.2: If... Then Reality Check Instructions: 1. You will be given five if-then statements, shown one after the other.	Activity A.2: If... Then Reality Check This activity introduces learners to simple conditional statements by asking them to judge whether they are true or false based on intuition. While intuition may work for simple statements with an obvious cause-and-

	• If a number is even, then it is divisible by 2. • If you are a student, then you must wear your school ID. • If the barangay captain declares a clean-up drive, then all residents will participate. • If the traffic light is red, then vehicles must stop. • If a person is below 18 years old, then he or she can vote in the national elections. 2. In each statement, evaluate whether it is true or false. 3. Given the LIKE emoji, orient it upwards (👍) if you think the statement is TRUE and downwards (👎) if it is false. 4. For every statement, one student will explain his/her response. Processing Questions: 1. How did you decide whether each statement was true or false? 2. Were there any statements that seemed confusing or difficult to judge? Why? 3. Do you think all if-then statements can be judged correctly using only intuition and common sense? Why or why not?	effect relationship between the <i>if</i> clause and the <i>then</i> clause, it is not always reliable in logical reasoning. In logic, a conditional or its forms does not require a cause-and-effect connection to be valid. This gap highlights the need for a systematic method which learners will develop in this lesson by using symbols and truth tables to evaluate conditional statements accurately. To facilitate this activity, follow the following steps: 1. Prepare five conditional statements, some obviously true and some clearly false. Each of these statements will be in a PowerPoint slide. You may consider using these statements below. • If a number is even, then it is divisible by 2. • If you are a student, then you must wear your school ID. 👍 • If the barangay captain declares a clean-up drive, then all residents will participate. 👍 • If the traffic light is red, then vehicles must stop. 👍 • If a person is 18 years old, then he or she can vote in the national elections. 👎 2. Provide a printed like emoji for each student that they can use to determine if the statement is true or false. In cases where printing is not accessible, let the students use their hands. 3. For every statement, ask one student to explain his answer. Ask other students if they have the opposite answer to foster critical thinking. 4. Ask students processing questions as suggested and connect their response of the third question to the need for a systematic approach in judging the truth value of a conditional statement and its forms which will be discussed in the lesson. Do it by saying: <i>“Not all statements can be judged correctly by intuition alone. That is why, we need a systematic method to evaluate conditional</i>
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		statements. In this lesson, we will learn how to construct related forms of conditional statements and how to evaluate its truth value.” 5. Present the Content and Performance Standards, the lesson objectives, and presentation of the unit performance task.							
B. Instituting New Knowledge	B.1. Presenting Examples								
	Activity B.1: Switch It, Flip It, Change It! 1. Consider the conditional proposition: <i>If the weather is good, then I will play outside.</i> 2. Let p: <i>The weather is good.</i> (hypothesis) q: <i>I will play outside.</i> (conclusion) 3. Your task is to create three new statements based on the original conditional by arranging the strips in the correct order following the instructions A, B, and C. 4. Create three new statements based on the original conditional by arranging and pasting the word strips in the second column as instructed in the first column. 5. Arrange and paste the symbol strips to its corresponding row in the third column. 6. After 5 minutes, you will paste your completed chart on the wall and ask 2 groups to present their output.	Activity B.1: Switch It, Flip It, Change It! This activity allows learners to actively manipulate conditional statements by switching and negating their parts, helping them observe how meanings shift. It provides an intuitive foundation for the formal definitions of converse, inverse, and contrapositive, which will be discussed in the next part of the lesson. By experiencing the transformations themselves, students better understand why these forms exist and how they relate to logical reasoning. Strategy 1: Strip Matching 1. Prepare two sets of strips, one set in words, and another set in symbols. Word Strips (the negated statement must have two strips each: <table><tr><td>the weather is good</td><td>the weather is not good</td></tr><tr><td>I will play outside</td><td>I will not play outside</td></tr></table> Symbol Strips (the negated symbols must have two strips each) <table><tr><td>P</td><td>$\sim P$</td></tr><tr><td>Q</td><td>$\sim Q$</td></tr></table>	the weather is good	the weather is not good	I will play outside	I will not play outside	P	$\sim P$	Q
the weather is good	the weather is not good								
I will play outside	I will not play outside								
P	$\sim P$								
Q	$\sim Q$								

	<table><tr><th>Instruction</th><th>Transformed Proposition</th><th>Symbolic Representation</th></tr><tr><td>A. Switch the hypothesis and the conclusion</td><td></td><td></td></tr><tr><td>B. Negate both the hypothesis and the conclusion</td><td></td><td></td></tr><tr><td>C. Switch and negate both the hypothesis and conclusion</td><td></td><td></td></tr></table>	Instruction	Transformed Proposition	Symbolic Representation	A. Switch the hypothesis and the conclusion			B. Negate both the hypothesis and the conclusion			C. Switch and negate both the hypothesis and conclusion			2. Prepare the chart with the table printed on it where the learners will paste the strips. The “If”, “then”, and the symbol $\rightarrow$ must be present already in the chart. <table><tr><th>Instruction</th><th>Transformed Proposition</th><th>Symbolic Representation</th></tr><tr><td>A. Switch the hypothesis and the conclusion</td><td>If I will play outside, then the weather is good.</td><td>$Q \rightarrow P$</td></tr><tr><td>B. Negate both the hypothesis and the conclusion</td><td>If the weather is not good, then I will not play outside.</td><td>$\sim P \rightarrow \sim Q$</td></tr><tr><td>C. Switch and negate both the hypothesis and conclusion</td><td>If I will not play outside, then the weather is not good.</td><td>$\sim Q \rightarrow \sim P$</td></tr></table>	Instruction	Transformed Proposition	Symbolic Representation	A. Switch the hypothesis and the conclusion	If I will play outside, then the weather is good.	$Q \rightarrow P$	B. Negate both the hypothesis and the conclusion	If the weather is not good, then I will not play outside.	$\sim P \rightarrow \sim Q$	C. Switch and negate both the hypothesis and conclusion	If I will not play outside, then the weather is not good.	$\sim Q \rightarrow \sim P$
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	Process Questions: - How did switching or negating the hypothesis and conclusion affect the original conditional statement? - What similarities or differences did you observe among the transformed statements? - Were there any transformations that seemed to keep the meaning of the original statement? Which ones? Why? How did the symbolic representations help you understand the changes in the statements?		- Divide the class into groups of 5 members. Each group will receive two sets of strips and the chart. - Project the given conditional statement on the screen and set the hypothesis as p and the conclusion as q. - Direct the groups to arrange strips for Row A, B, and C. - As the groups perform the task, go around one group to the next to check their work and to give clarifications. - Ask one or two groups to present their final strip arrangements to the class for validation.																							

		8. Ask the class a series of processing questions. Conclude by noting that these three transformations have special names (to be introduced in Part B2).
B.2. Discussing the Concept		
	Activity B.2: Lesson Discussion: Forms of Conditional Propositions Instructions: To be able to understand the key concepts of the lesson, listen attentively at participate actively to the discussion. Conditional Proposition A conditional proposition has two parts: 1. A hypothesis (or antecedent), denoted by P 2. A conclusion (or consequent), denoted by Q In symbols, the statement, “If P, then Q” is written as $P \rightarrow Q$. Example: If <i>the weather is good</i>, then <i>I will play outside</i>. $\qquad\qquad\qquad P \qquad\qquad\qquad Q$ Forms of Conditional Statement 1. Converse The converse of a conditional proposition is formed by reversing the hypothesis and the conclusion. In symbols, the statement “If Q, then P” is written as: $Q \rightarrow P$. Example: If <i>I will play outside</i>, then <i>the weather is good</i>. $\qquad\qquad\qquad Q \qquad\qquad\qquad P$	Activity B.2: Lesson Discussion: Forms of Conditional Propositions To deepen the learners’ understanding of the forms of conditional propositions, facilitate the discussion of the topic using the following steps: 1. Organize groups in the class composed of three members. 2. Provide a handout that explains the key concepts in the lesson. 3. Provide examples of conditional statements which the students will identify first the hypothesis and the conclusion. 4. In introducing the forms of conditional propositions, use the statements they have arranged in the earlier activity as examples. This will foster the feeling of discovery among students. 5. As the lesson progresses, use the examples in Step 3 to call multiple groups, with each group will recite one form of the given conditional proposition. This makes the discussion an active engagement because the students not only listen but share ideas in the class. 6. Encourage peer feedback as different groups share their answer for increased engagement. You may ask them: “Do you agree with this contrapositive? Why or why not?”

2. Inverse

The inverse of a conditional proposition is formed by **negating both the hypothesis and the conclusion**. It is denoted in symbols as: $\sim P \rightarrow \sim Q$.

Example:

If *the weather is not good*, then *I will not play outside*.
 $\sim P$ $\sim Q$

3. Contrapositive

The contrapositive of a conditional proposition is formed by **reversing and negating both the hypothesis and the conclusion**. It is denoted in symbols as: $\neg Q \rightarrow \neg P$

Example:

If *I will not play outside*, then *the weather is not good*.
 $\sim Q$ $\sim P$

Truth Value of Conditional Propositions

A conditional proposition may be true or false. To show that a conditional proposition is false, you only need to find one example (called a **counter example**) in which the hypothesis is fulfilled, and the conclusion is not fulfilled. For example, in the conditional statement: *If the weather is good, then I will play outside*, it can be interpreted as false **when the weather is good, but I will not play outside**.

The valuation of a conditional proposition can be described using a truth table.

P	Q	$P \rightarrow Q$
T	T	T

7. If the class has access to the internet, you may adjust the activity to let every group construct the forms of conditional propositions using PADLET.

T	F	F
F	T	T
F	F	T

Example:

Suppose that Mark is a Grade 12 student. Consider this conditional proposition:

Proposition:

If Mark is in Grade 12, then he is a Senior High School student. ($P \rightarrow Q$)

What is the truth value of the proposition?

Solution:

“Mark is in Grade 12” and “Mark is a senior high school student” are both true because Grade 12 is part of senior high school. Since both the hypothesis and conclusion are true, then the proposition is TRUE, based on the first row of the truth table.

What is the converse, inverse, and contrapositive of the conditional proposition? Express in words and symbols.

Solution:

1. Converse: If Mark is a senior high school student, then he is in Grade 12. ($Q \rightarrow P$)
2. Inverse: If Mark is not in Grade 12, then he is not a senior high school student. ($\sim P \rightarrow \sim Q$)
3. Contrapositive: If Mark is not a senior high school student, then he is not in Grade 12. ($\sim Q \rightarrow \sim P$)

What is the truth value of each form of the conditional statement?

	Solution: 1. Converse: Both the hypothesis and conclusion are true, so the converse of the conditional proposition is also TRUE. 2. Inverse: “Mark is not in Grade 12” and “Mark is not a senior high school” are both false since it was stated that he is in Grade 12. Since both the hypothesis and conclusion is false, the inverse is TRUE as shown in the fourth row of the truth table. 3. Contrapositive: “Mark is not a senior high school” and “Mark is not in Grade 12” are both false since it was stated that he is in Grade 12. Since both the hypothesis and conclusion are false, the contrapositive is TRUE as shown in the fourth row of the truth table.	
	B.3. Developing Mastery	
	Activity B.3: Transform It! Construct the converse, inverse, and contrapositive of the following conditional propositions in words and symbols. 1. If a person is a teacher, then the person prepares lessons. Converse: _____ Inverse: _____ Contrapositive: _____ 2. If today is a weekday, then students go to school. Converse: _____ Inverse: _____ Contrapositive: _____	Activity B.3: Transform It! This activity gives learners structured practice in applying what they learned. By repeatedly transforming conditional statements into their converse, inverse, and contrapositive, they strengthen their ability to work systematically with both sentence form and symbols. This mastery is essential before moving on to analyzing truth values later in the lesson. To facilitate this activity, make the following preparations: 1. Prepare a worksheet with two conditional propositions. ● If a person is a teacher, then the person prepares lessons. Solution:

		<i>Converse: If the person prepares lessons, then the person is a teacher. ($Q \rightarrow P$)</i> <i>Inverse: If a person is not a teacher, then the person does not prepare lessons. ($\sim P \rightarrow \sim Q$)</i> <i>Contrapositive: If the person does not prepare lessons, then the person is not a teacher. ($\sim Q \rightarrow \sim P$)</i> • If today is a weekday, then students go to school. Solution: <i>Converse: If students go to school, then today is a weekday. ($Q \rightarrow P$)</i> <i>Inverse: If today is not a weekday, then students do not go to school. ($\sim P \rightarrow \sim Q$)</i> <i>Contrapositive: If students do not go to school, then today is not a weekday. ($\sim Q \rightarrow \sim P$)</i> 2. In each conditional statement, provide a space for its converse, inverse, and contrapositive. Strategy 1: Individual Work 1. Distribute the worksheet. 2. Allow students to work silently for 5 minutes. 3. Afterwards, discuss answers as a class, highlighting common errors. Strategy 2: Group Challenge 1. Divide class into small groups (3–4 members). 2. Assign the same worksheet. 3. Groups work together to complete the transformations.
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		Turn it into a “race” — first three groups to finish correctly earn points or small recognition.
D. Demonstrating Knowledge and Skills	C.1. Finding Practical Application	
	Activity C.1: Fact or Bluff? A food supplement brand is heavily promoting its product through their sellers. They claim that: “If you take a tablet of X food supplement, then your illnesses will be healed.” At first glance, the claim sounds convincing and hopeful, especially for people looking for an easy cure. However, on the product’s packaging, it clearly states: “No Approved Therapeutic Claims.” In addition, several customers have reported that their illnesses did not improve even after taking the supplement. This raises the question: Is the food supplement brand’s conditional claim true and valid? Task: Write a short 6–10 line conversation among three to four people in which they: 1. Restate the claim in a conditional statement. 2. Judge whether the claim is true or false in this context, using the truth table and the evidence given. 3. Explain why you consider the claim valid or invalid, based on the packaging disclaimer and customer reports as supporting evidence. 4. Present the dialogue in the class for 3 minutes per group.	Activity C.1: Fact or Bluff? This activity helps learners critically assess real-world conditional claims about products promoted by sellers. Through a group conversation supported by truth-table reasoning and other evidences such as label disclaimers and customer feedback, they develop logical thinking, media literacy, and informed decision-making skills as discerning consumers. To facilitate this activity, follow the following steps: 1. Present the scenario about the seller promoting Brand X’s food supplement to the customer. Distribute or project the claim and the additional context (label disclaimer and customer feedback). 2. Instruct the class to form groups of three to four members. Each group will create a short conversation showing the claim of the seller, and how they analyze the truth and validity of the sellers’ claim. The conversation should include: • Restating the claim in a conditional statement; • Judging whether the claim is true or false based on the context and concept from the truth table; and • Explaining their reasoning with logical evidence. 3. After group work, invite the groups to present their conversations to the class. Facilitate a short

Note: The cast of the conversation will have one seller, one customer, and one or two former customers who already used the product.

Criteria	Excellent	Good	Satisfactory	Needs Improvement
Content Accuracy and Relevance (12 points)	11–12 pts Clear, accurate restatement; correct truth judgement; strongly supported by evidence.	8–10 pts Accurate restatement; correct judgement but explanation partly lacking.	5–7 pts Minor errors or weak link to context.	0–4 pts Inaccurate restatement or unsupported/incorrect judgement.
Reasoning and Use of Truth Table (4 points)	4 pts Applies truth-table reasoning correctly; clear logical explanation.	3 pts Mostly correct use; minor lapses in explanation.	2 pts Some attempt but misapplied or unclear logic.	0–1 pt No correct reasoning or truth-table use.
Conversation Delivery and Interaction (4 points)	4 pts Conversation flows naturally among the members; each speaker contributes meaningfully and responds to others' ideas; delivery is clear and engaging.	3 pts All members participate; interaction is generally smooth though some turns may be uneven or less connected.	2 pts Limited interaction; some members dominate or contribute minimally; conversation lacks coherence or energy.	0–1 pt Minimal participation or interaction; conversation is unclear, incomplete, or lacks collaboration.
Total (20 points)				

discussion comparing their analyses and highlighting correct logical reasoning and evidence use.

4. Use the succeeding rubric to rate their performance.

Criteria	Excellent	Good	Satisfactory	Needs Improvement
Content Accuracy and Relevance (12 points)	11–12 pts Clear, accurate restatement; correct truth judgement; strongly supported by evidence.	8–10 pts Accurate restatement; correct judgement but explanation partly lacking.	5–7 pts Minor errors or weak link to context.	0–4 pts Inaccurate restatement or unsupported/incorrect judgement.
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Total (20 points)				

C.2. Making Generalization

Activity C.2: 3-2-1 Exit Card

On a sheet of paper, write your responses to the following prompts before we wrap up the lesson:

3 things you learned about conditional statements and their forms.

Activity C.2: 3-2-1 Exit Card

This strategy is efficient and highly reflective. It helps learners process the key takeaways from the lesson in a structured way while giving the teacher quick feedback on the class's understanding.

To facilitate this activity smoothly, follow these steps:

1. Distribute index cards or ask students to use half a sheet of paper.

	2 ways to evaluate if a conditional statement or its forms is true or false. 1 reason why systematic evaluation (using logic) is better than relying only on intuition.	2. Display the 3-2-1 prompts on the board. 3. Give students 3–5 minutes to complete their cards individually. 4. Collect the cards to quickly assess whether learners grasped the key concepts. End the session with a short synthesis, highlighting recurring points from their responses.
	C.3. Evaluating Learning	
	Activity C.3: Logic Checkpoint Instructions: Choose the letter of the correct answer. Write your answers on a sheet of paper. 1. Which of the following is the converse of the conditional statement “<i>If it rains, then the ground will be wet</i>”? A. If it does not rain, then the ground will not be wet. B. If the ground is wet, then it rains. C. If the ground is wet, then it rained. D. If it rains, then the ground is not wet. 2. The statement “<i>If a person is a senior citizen, then they are at least 60 years old</i>” is: A. True B. False C. Sometimes true, sometimes false D. Cannot be determined 3. What is the inverse of the statement “<i>If a student studies well, then they will pass the exam</i>”? A. If a student does not study well, then they will pass the exam.	Activity C.3: Logic Checkpoint This quiz is designed to assess learners’ understanding of conditional statements, including their converse, inverse, and contrapositive forms, as well as their ability to evaluate truth values. The activity ensures that students can apply logical reasoning not just in mathematics but also in analyzing real-life claims. The quiz may be administered in printed form for face-to-face classes or through online platforms such as Kahoot! to make the assessment interactive and engaging. Learners answer individually to measure mastery, but results can be reviewed as a class to address common misconceptions. Answer Key: 1. B 2. A 3. B 4. A 5. C

	B. If a student does not study well, then they will not pass the exam. C. If a student passes the exam, then they studied well. D. If a student does not pass the exam, then they did not study well. 4. Which of the following is the contrapositive of the statement “If a product is safe, then it has passed quality testing”? A. If a product does not pass quality testing, then it is not safe. B. If a product is not safe, then it has passed quality testing. C. If a product is safe, then it did not pass quality testing. D. If a product passes quality testing, then it is safe. 5. If the statement “If P, then Q” is given, which of the following correctly shows its contrapositive in symbols? A. $\sim P \rightarrow \sim Q$ B. $Q \rightarrow P$ C. $\sim Q \rightarrow \sim P$ D. $P \rightarrow \sim Q$	
C.4. Additional Activities		
	Activity C.4: Additional Exercise in Constructing Forms of Conditional Propositions Remediation Construct the converse, inverse, and contrapositive of the given conditional propositions. 1. If a number is divisible by 4, then it is even.	Activity C.4: Additional Exercise in Constructing Forms of Conditional Propositions This activity strengthens the learners’ ability to construct the converse, inverse, and contrapositive of conditional statements. By offering remediation and enhancement tasks, the teacher ensures that all

	Converse: _____ Inverse: _____ Contrapositive: _____ 2. If a polygon has four equal sides, then it is a square. Converse: _____ Inverse: _____ Contrapositive: _____ 3. If a person studies well, then they will pass the exam. Converse: _____ Inverse: _____ Contrapositive: _____ Enhancement Create your one original conditional statement on each of the following topics: 1. School rules (ex: If you are late, then ...) 2. Community or law (ex: If you litter, then ...) 3. Mathematics (ex: If a polygon has 3 sides, then ...) For each conditional statement you make, write its converse, inverse, and contrapositive.	learners—whether struggling or advanced—are supported and challenged appropriately.
III. CONTENT	Lesson 12.3. Syllogisms: Modus Ponens and Modus Tollens, Inverse and Converse Fallacy Recommended time allotment: 2 hours	
IV. OBJECTIVES	At the end of the lesson, learners should be able to: 1. define syllogisms such as <i>Modus Ponens</i> and <i>Modus Tollens</i>, and fallacies like the <i>inverse</i> and <i>converse fallacy</i>. 2. differentiate syllogisms such as <i>Modus Ponens</i> and <i>Modus Tollens</i> from fallacies like the <i>inverse</i> and <i>converse fallacy</i>.	

	3. illustrate tautologies and contradictions through real-life examples and practical situations. 4. determine the truth values of propositions in given statements using logical reasoning. 5. apply Modus Ponens and Modus Tollens correctly in analyzing day-to-day arguments or scenarios. 6. identify and explain instances of inverse and converse fallacies in practical situations. 7. evaluate the validity of arguments by checking for tautologies, contradictions, and fallacies.	
V. PROCEDURES	ACTIVITIES (Instructions for Learners)	ACTIVITIES (Instructions for Teachers)
A. Activating Prior Knowledge	A. Eliciting Prior Knowledge	
	Activity: WHAT’S THE CONCLUSION? This activity engages you to apply your prior knowledge of conditional or <i>if-then</i> statements through practical examples that mirror real-life situations. You will work collaboratively in groups. Instructions: The class will be divided into small groups. Groups of 4-6 members Each group will work together on 5 cases given by the teacher on the board, projector, or handout. Example A: <i>If you submit your project on time, then you will get a grade. You submitted your project on time.</i> → Conclusion: You will get a grade. Decide if a valid conclusion can be made. ❖ If yes, write the conclusion. ❖ If no, explain why no valid conclusion is possible. Record your group’s answer on the worksheet.	Activity: WHAT’S THE CONCLUSION? This activity aims to engage learners in applying their prior knowledge of conditional or <i>if-then</i> statements through practical examples that mirror real-life situations. By working collaboratively in groups, students will develop their critical thinking skills as they analyze whether a given conclusion logically follows from the premises provided. Strategy 1: Group Analysis of Conditional Statements Purpose: Help students recall and apply their prior knowledge of conditional (<i>if-then</i>) statements in practical situations to strengthen their critical thinking by identifying valid arguments and distinguishing them from fallacies. Materials: ✓ Printed worksheets (with cases and spaces for group answers), ✓ Pen or pencil (for writing group responses) ✓ Chalkboard/whiteboard or projector (for group sharing and discussion) ✓ Timer Instructions:

	5. You will assign a spokesperson to present your answers and reasoning to the class. Process Questions: 1. What conclusion did your group come up with in Case 1? How did you know it follows from the given statements? 2. In Case 2, can we really say anything certain about the situation? Why or why not? 3. In Case 3, does the conclusion always happen in real life, or is it just based on the way the statements are written? 4. In Case 4, what made you decide on your conclusion? What was the reasoning behind it? 5. In Case 5, why do you think we cannot be fully sure of the conclusion? What other possibilities might exist? 6. Looking at all the cases, which ones felt more “certain” or stronger in reasoning? Which ones felt weaker or doubtful?	To facilitate this activity, follow these steps: 1. Begin by arranging the class into groups of 4-6 members. 2. Remind students about conditional (<i>if-then</i>) statements. Give a simple example to refresh their memory. 3. Provide a worksheet with 5 cases and space for group answers. Have a copy of the answer key for debriefing. Case 1 Statement: If you bring an umbrella, then you won’t get wet. You brought an umbrella. Conclusion: You won’t get wet. Case 2 Statement: If it is Saturday, then there is no class. Today is not Saturday. Conclusion: No valid conclusion. Case 3 Statement: If you play basketball every day, then you will get taller. Juan plays basketball every day. Conclusion: Juan will get taller. Case 4 Statement: If you are a teacher, then you can explain lessons. Anna cannot explain the lessons. Conclusion: Anna is not a teacher. Case 5 Statement: If it rains, then the ground gets wet. The ground is wet. Conclusion: It rained.
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		4. Ask the students to work together to solve the 5 cases. 5. Encourage each group to justify their answers. 6. Call each group’s spokesperson to share answers to at least one case. 7. Compare group responses; allow clarifications. 8. Go through each case with the whole class. a. Highlight which are valid and which are fallacies b. Emphasize the importance of correct reasoning in everyday decision-making. Process Questions: 1. What conclusion did your group come up with in Case 1? How did you know it follows from the given statements? (Encourages learners to explain their reasoning and justify why a conclusion can be made.) 2. In Case 2, can we really say anything certain about the situation? Why or why not? (Helps learners recognize situations where no valid conclusion can be drawn.) 3. In Case 3, does the conclusion always happen in real life, or is it just based on the way the statements are written? (Prompts learners to distinguish between logical validity and real-life truth.) 4. In Case 4, what made you decide on your conclusion? What was the reasoning behind it? (Encourages learners to trace the reasoning process and explain how they ruled something out.) 5. In Case 5, why do you think we cannot be fully sure of the conclusion? What other possibilities might exist? (Guides learners to see the limitations of
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		<i>reasoning and the possibility of alternative explanations.)</i> 6. Looking at all the cases, which ones felt more “certain” or stronger in reasoning? Which ones felt weaker or doubtful? <i>(Helps learners compare arguments and reflect on the strength or weakness of reasoning.)</i> Strategy 2: “Interactive Gallery Walk” Purpose: More engaging and interactive activity by having learners evaluate and compare reasoning across groups, fostering peer critique, and collaborative reflection. Materials: ✓ Prepared list of conditional statements and possible conclusions ✓ Manila paper/ cartolina or large sheets ✓ Markers (for writing group responses) ✓ Masking Tape (to post answers on the wall/board) ✓ Timer Instructions: To facilitate this activity, follow these steps: 1. Begin by arranging the class into groups of 4-6 members. 2. Remind students about conditional (<i>if-then</i>) statements. Give a simple example to refresh their memory. 3. Provide a worksheet with 5 cases and space for group answers. Have a copy of the answer key for debriefing.
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		9. Link to the importance of identifying valid versus invalid reasoning in logic.
	A.2. Establishing the Purpose of the Lesson	
	Activity: RIGHT OR WRONG DECISION? This activity is presented through short, real-life scenarios involving decisions based on “if-then” reasoning. They will judge whether the conclusion made is reasonable or faulty. This activity sets the stage for understanding the importance of learning correct reasoning patterns and avoiding fallacies. Procedures: 1. Read each scenario carefully. 2. Work in pairs or groups of 3. 3. Decide: Is the conclusion right (valid) or wrong (invalid/uncertain)? 4. Explain your reasoning briefly. 5. Be ready to share one example with the class. Scenario 1: If you wake up early, then you will catch the school bus. You woke up early. → You will catch the school bus. Scenario 2: If it rains, then the ground is wet. The ground is wet. → Therefore, it rained. Scenario 3: If you eat healthy food, then you will stay strong. Ana is strong. → Therefore, Ana eats healthy food. Scenario 4: If a number is divisible by 10, then it ends with 0.	Activity: RIGHT OR WRONG DECISION? Strategy 1: Think-Pair-Share Purpose: To maximize individual reflection, peer discussion, and whole-class sharing, ensuring all learners engage with the reasoning process. Materials: ✓ Pre-made scenario cards ✓ Clipboards or desk space ✓ Timer ✓ Sharing rubric or checklist (optional) ✓ Visual cue cards for scaffolding (optional) Instructions: 1. Distribute 3–4 sentence cards to each learner. Each card contains a different scenario. Scenario 1: If you wake up early, then you will catch the school bus. You woke up early. → You will catch the school bus. Scenario 2: If it rains, then the ground is wet. The ground is wet. → Therefore, it rained.

45 ends with 5.
→ Therefore, it is divisible by 10.

Scenario 5:

If a person studies well, then they will pass the exam.
Juan passed the exam.
→ Therefore, Juan studied well.

Scenario 6:

If the electricity is off, then the lights will not work.
The lights are not working.
→ Therefore, the electricity is off.

Scenario 7:

If a shape is a square, then it has four sides.
This figure is a square.
→ Therefore, it has four sides.

Scenario 8:

If an animal is a dog, then it is a mammal.
A cat is a mammal.
→ Therefore, a cat is a dog.

Scenario 9:

If you practice basketball daily, then you will improve your skills.
Maria improved her basketball skills.
→ Therefore, Maria practiced daily.

Scenario 10:

If a plant does not get sunlight, then it will not grow well.
This plant did not grow well.
→ Therefore, it did not get sunlight.

Process Questions:

Scenario 3:

If you eat healthy food, then you will stay strong.
Ana is strong.
→ Therefore, Ana eats healthy food.

Scenario 4:

If a number is divisible by 10, then it ends with 0.
45 ends with 5.
→ Therefore, it is divisible by 10.

Scenario 5:

If a person studies well, then they will pass the exam.
Juan passed the exam.
→ Therefore, Juan studied well.

Scenario 6:

If the electricity is off, then the lights will not work.
The lights are not working.
→ Therefore, the electricity is off.

Scenario 7:

If a shape is a square, then it has four sides.
This figure is a square.
→ Therefore, it has four sides.

Scenario 8:

If an animal is a dog, then it is a mammal.
A cat is a mammal.
→ Therefore, a cat is a dog.

Scenario 9:

If you practice basketball daily, then you will improve your skills.
Maria improved her basketball skills.
→ Therefore, Maria practiced daily.

	1. Which scenarios had strong and certain reasoning? 2. Which ones seemed doubtful or misleading? Why? 3. How can making the wrong conclusion affect us in real life? 4. Why do you think we need to study logical rules before making judgments?	Scenario 10: If a plant does not get sunlight, then it will not grow well. This plant did not grow well. → Therefore, it did not get sunlight. Valid reasoning (Modus Ponens / Modus Tollens): Scenarios 1, 7 Fallacy of affirming the consequent: Scenarios 2, 5, 9, 10 Fallacy of denying the antecedent: Scenarios 3, 4, 6, 8 2. The learners will discuss their answers with their partners, justifying their reasoning. 3. Call on several pairs to share their conclusions. Record their answers on the board (✓ for valid, ✗ for invalid/uncertain). 4. Ask: “What patterns do you notice in valid reasoning? What mistakes appeared in the wrong ones?” → Transition into the purpose of the lesson. Process Questions: 1. Which scenarios had strong and certain reasoning? <i>(Helps learners sense Modus Ponens & Modus Tollens intuitively.)</i> 2. Which ones seemed doubtful or misleading? Why? <i>(Leads them to see common fallacies.)</i> 3. How can making the wrong conclusion affect us in real life? <i>(Links lesson to practical decision-making.)</i> 4. Why do you think we need to study logical rules before making judgments? <i>(Establishes lesson purpose.)</i>
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Strategy 2: Four Corners Debate

Purpose:

To encourage movement, active participation, and critical thinking through group defense of reasoning.

Materials:

- ✓ Labels for the four classroom corners (**Right, Wrong, Uncertain, Needs Discussion**).
- ✓ List of Scenarios

Instructions:

1. Label the four classroom corners: **Right, Wrong, Uncertain, Needs Discussion**.
2. Read one scenario aloud. Ask groups to move to the corner that represents their judgment of the conclusion.

Scenario 1:

If you wake up early, then you will catch the school bus.

You woke up early.

→ You will catch the school bus.

Scenario 2:

If it rains, then the ground is wet.

The ground is wet.

→ Therefore, it rained.

Scenario 3:

If you eat healthy food, then you will stay strong.

Ana is strong.

→ Therefore, Ana eats healthy food.

		Scenario 4: If a number is divisible by 10, then it ends with 0. 45 ends with 5. → Therefore, it is divisible by 10. Scenario 5: If a person studies well, then they will pass the exam. Juan passed the exam. → Therefore, Juan studied well. Scenario 6: If the electricity is off, then the lights will not work. The lights are not working. → Therefore, the electricity is off. Scenario 7: If a shape is a square, then it has four sides. This figure is a square. → Therefore, it has four sides. Scenario 8: If an animal is a dog, then it is a mammal. A cat is a mammal. → Therefore, a cat is a dog. Scenario 9: If you practice basketball daily, then you will improve your skills. Maria improved her basketball skills. → Therefore, Maria practiced daily. Scenario 10: If a plant does not get sunlight, then it will not grow well.
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		This plant did not grow well. → Therefore, it did not get sunlight. Valid reasoning (Modus Ponens / Modus Tollens): Scenarios 1, 7 Fallacy of affirming the consequent: Scenarios 2, 5, 9, 10 Fallacy of denying the antecedent: Scenarios 3, 4, 6, 8 3. Call on each corner to justify their choice. Encourage rebuttals from other corners. 4. After discussion, reveal whether the reasoning is valid or invalid, explaining why. 5. Repeat for all scenarios. End with a synthesis: <i>“Some reasoning feels convincing, but it is not logically valid. That’s why today we’ll study rules to check validity.”</i> <i>“These follow-up questions serve as a bridge to the performance task that deepens learners’ understanding of logical propositions and truth values in Mathematics”</i>
B. Instituting New Knowledge	B.2. Discussing the Concept	
	Activity B.2: Deepening Understanding on Propositions Instructions: To be able to understand the key concepts of the lesson, listen attentively at participate actively to the discussion TAUTOLOGY Definition:	Activity B.2: Deepening Understanding on Propositions Strategy 1: Truth Table Exploration (Inquiry-Based / Constructivist Approach) Objective: To help students understand tautology by discovering that a tautological statement is always <i>true</i>, regardless of the truth values of its variables. Steps:

A proposition that is always true is called a **tautology**, a proposition that is always false, is called a **contradiction**.

A **tautology** is a propositional form that is always true regardless of the truth values of the individual statements substituted for its propositional variables. The word tautology is derived from a Greek word where ‘tauto’ means ‘same’ and ‘logy’ means ‘logic’.

A proposition whose form is a tautology is called a tautological statement. A truth table shows how the truth or falsity of a compound statement depends on the truth or falsity of the simple statements from which it's constructed.

The simple examples of tautology are:

- Either Mari will buy apples or Mari will not buy apples.
- My pet Yummy is healthy, or he is not healthy
- A function is a polynomial function or it is not a polynomial function.

TYPES OF TAUTOLOGIES:

Let p, q, and r be propositions.

Example 1. Prove that the argument $[(p \rightarrow q) \wedge p] \rightarrow q$ is valid. This argument is known as **Modus Ponens** (or Rule of Detachment).

Solution. We only need to show that $[(p \rightarrow q) \wedge p] \rightarrow q$ is a tautology.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$[(p \rightarrow q) \wedge p] \rightarrow q$
T	T	T	T	T

1. **Start with a Review:** Review basic logical connectives ($\neg$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$) and how they work in truth tables.

2. **Introduce Example:** Present sample statements such as:

- $p \vee \neg p$
- $(p \rightarrow q) \vee (q \rightarrow p)$

3. **Construct the Truth Table Together:** Have students fill out the table for each case. Let them observe that in every row, the statement is **true**.

4. **Discuss the Result:** Explain that because the compound statement is *always true*, it is called a **tautology**.

5. **Extension:**

Ask students to create their own examples of tautological statements and verify them using truth tables.

Materials Needed: Whiteboard, projector, or printed truth table templates.

Strategy 2: Real-Life Analogies and Logical Reasoning (Contextual/ Experiential Approach)

Objective: To connect tautology to everyday reasoning and enhance conceptual understanding.

Steps:

1. **Relate to Everyday Situations:** Start with simple examples like:
 - “Either it will rain tomorrow, or it won’t.”
 - “All bachelors are unmarried men.”
 These statements are *always true by*

T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since $[(p \rightarrow q) \wedge p] \rightarrow q$ is a tautology, then the argument is **valid**.

Example 2. Consider the arguments below:

Argument A

If my alarm sounds, then I will wake up.
My alarm sounded.
Therefore, I woke up.

Argument B

If there is limited freshwater supply, then we should conserve water.
There is limited water supply.
Therefore, we should conserve water.

Argument C

If General Antonio Luna is a national hero, then he died at the hands of the Americans in 1899.
General Antonio Luna is a national hero.
Therefore, General Antonio Luna died at the hands of the Americans in 1899.

Solution. Note that they are all of the form $[(p \rightarrow q) \wedge p] \rightarrow q$, or in standard form

$$\begin{array}{r} p \rightarrow q \\ p \\ \hline \therefore q \end{array}$$

Hence, by **Modus Ponens**, arguments A, B, and C are all valid. However, this does not mean that the conclusions are true. Asserting that an argument is valid means that the conclusion logically follows the premise.

definition—helping students intuitively grasp tautology.

2. **Introduce Symbolic Form:** Convert everyday examples into symbolic logic (e.g., $p \vee \neg p$) and explain that it expresses the same truth in formal logic.
3. **Group Activity:** Divide students into small groups and give each group real-life statements. Ask them to decide whether each is a tautology, contradiction, or contingency—and explain their reasoning.
4. **Class Discussion:** Groups share their answers, and you guide the class to refine their understanding through reasoning and examples.

Materials Needed: Handouts with sample statements, marker boards, or digital slides.

Example 3.

If I follow the school rules and regulations, then I am a disciplined person, but I follow the school rules and regulations, therefore I am a disciplined person.

Show that the given statement is a tautology.

Solution:

Step 1: Translate the given statement into symbols.

Let p : I follow school rules and regulations.

q : I am a disciplined person.

The statement can be written in symbols as

$$[(p \rightarrow q) \wedge p] \rightarrow q.$$

Step 2: Construct the truth table of the given statements.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$[(p \rightarrow q) \wedge p] \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Step 3: Make a conclusion.

Since the result is all true, it shows that the given statement is a tautology. Observe also that the statement illustrates **Modus Ponens**.

Example 4. (Conversation)**Family Scenario — Modus Ponens**

Mom: If I cook adobo tonight, we'll have dinner early.

You: Oh, you're cooking adobo?

Note: *“If the teacher wishes to include additional conversation examples, she may provide them.”*

Mom: Yes!

You: Then we'll have dinner early!

☒ **Type: *Modus Ponens***

✱ If $P \rightarrow Q$; P ; therefore Q .

(If Mom cooks adobo, then we'll eat early. Mom cooks adobo, therefore we'll eat early.)

Example 5.

If I study hard, then I will pass the exam.

I did not pass the exam; therefore, I did not study hard.

Show that the given statement is a tautology.

Solution:

Step 1: Translate the given statement into symbols.

Let p : I study hard.

q : I will pass the exam.

The statement can be written in symbols as:

$$(p \rightarrow q) \wedge \sim q \rightarrow \sim p.$$

Step 2: Construct the truth table of the given statements.

p	q	$p \rightarrow q$	$\sim q$	$(p \rightarrow q) \wedge \sim q$	$[(p \rightarrow q) \wedge \sim q] \rightarrow \sim p$
T	T	T	F	F	T
T	F	F	T	F	T
F	T	T	F	F	T
F	F	T	T	T	T

Step 3: Make a conclusion.

Since the result is all true, it shows that the given statement is a tautology. Observe also that the statement illustrates **Modus Tollens**.

Example 6. (Conversational)

Friendship and Trust — Modus Tollens

You: If Sarah really trusted me, she'd tell me her secret.

Friend: But she didn't tell you, right?

You: Yeah... so maybe she doesn't trust me after all.

☒ Type: *Modus Tollens*

✱ If $P \rightarrow Q$; $\sim Q$; therefore $\sim P$.

On the other hand, an argument that is not valid is called a **fallacy**. Hence, in a fallacy, the premises may be true, while the conclusion is false. In a truth table, we call such set as a counterexample

Consequently, the conditional $(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$ is not a tautology. For instance, argument G' is a fallacy as can be seen in the first truth table presented in this lesson. In standard form it is

$$\begin{array}{c} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

This argument is a **fallacy of the converse**.

FALLACY

Definition

An argument

$$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q$$

which is not valid is called a **fallacy**.

In a fallacy, it is possible for the premise $p_1, p_2, \dots, p_n$ to be true while the conclusion q is false.

Equivalently, for this case, the conditional

$$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \rightarrow q,$$

is not a tautology.

TABLE OF FALLACIES

Let p, q , and r be propositions.

RULES OF INFERENCE	PROPOSITIONAL FORM	STANDARD FORM
Fallacy of the Converse	$[(p \rightarrow q) \wedge q] \rightarrow p$	$\frac{p \rightarrow q}{q} \therefore p$
Fallacy of the Inverse	$[(p \rightarrow q) \wedge (\sim p)] \rightarrow \sim q$	$\frac{p}{\therefore p \vee q}$
Affirming the Disjunct	$[(p \vee q) \wedge p] \rightarrow \sim q$	$\frac{p \vee q}{p} \therefore \sim q$
Fallacy of the Consequent	$(p \rightarrow q) \rightarrow (q \rightarrow p)$	$\frac{p \rightarrow q}{\therefore q \rightarrow p}$
Denying a Conjunct	$[\sim(p \wedge q) \wedge (\sim p)] \rightarrow q$	$\frac{[\sim(p \wedge q)]}{\sim p} \therefore q$
Improper Transposition	$(p \rightarrow q) \rightarrow [(\sim p) \rightarrow (\sim q)]$	$\frac{p \rightarrow q}{\therefore (\sim p) \rightarrow (\sim q)}$

Example 5. Prove that the argument $[(p \rightarrow q) \wedge q] \rightarrow p$ is a fallacy. This is known as the **Fallacy of the Converse**.

Solution. We show that $[(p \rightarrow q) \wedge q] \rightarrow p$ is not a tautology by means of a truth table.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge q$	$[(p \rightarrow q) \wedge q] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Since $[(p \rightarrow q) \wedge p] \rightarrow p$ is not a tautology, the argument is invalid and is hence a **fallacy**. Note that it is sufficient to find truth values of **p** and **q** that will make the premises simultaneously true but the conclusion false. We call such a set of values a counterexample. In this case, the counterexample is when **p** is false and **q** is true (see the third row of the given truth table).

Example 7. Show that the following arguments are fallacies.

Argument A'

If my alarm sounds, then I will wake up.
I woke up.
Therefore, my alarm sounded.

Argument B'

If there is limited supply of freshwater, then I will conserve water.
I will conserve water.
Therefore, there is limited supply of freshwater.

Solution. Each of the arguments has the form

$$\begin{array}{c} p \rightarrow q \\ q \\ \hline \therefore p \end{array}$$

Thus, each argument is a **fallacy of the converse**.

For argument A', it is possible that I woke up because of a bad dream, or because I suddenly remembered that I still have to study for my logic exam.

For argument B', even if I will conserve fresh water, it does not logically imply that there is a limited supply of freshwater. Perhaps, I just want to lower my monthly bill!

Example 8. Prove that the argument $[(p \rightarrow q) \wedge \sim p] \rightarrow \sim q$ is a fallacy. This is known as the **Fallacy of the Inverse**.

Solution. We show that $[(p \rightarrow q) \wedge \sim p] \rightarrow \sim q$ is not a tautology by means of a truth table

p	q	$p \rightarrow q$	$\sim p$	$(p \rightarrow q) \wedge \sim p$	$[(p \rightarrow q) \wedge \sim p] \rightarrow \sim q$
T	T	T	F	T	T
T	F	F	F	F	T
F	T	T	T	T	F
F	F	T	T	F	T

Since $[(p \rightarrow q) \wedge \sim p] \rightarrow \sim q$ is not a tautology, the argument is invalid and is hence a **fallacy**

Note that it is sufficient to find truth values of **p** and **q** that will make the premises simultaneously true but the conclusion false.

We call such a set of values a **counterexample**.

In this case, the counterexample is when **p** is false and **q** is true (see the third row of the given truth table).

	B.3. Developing Mastery	
	Activity: “Logic Detectives” 🔍 Scenario for Students: You are now <i>Logic Detectives</i>. Each case contains a reasoning statement. Your mission is to analyze the case and decide whether the reasoning follows Modus Ponens, Modus Tollens, Fallacy of the Converse, or Fallacy of the Inverse. ☒ Valid- Modus Ponens and Modus Tollens ☒ Invalid- Fallacy of the Inverse and Fallacy of the Converse Instructions: 1. Read each case carefully. 2. Identify the reasoning pattern used. 3. Write the correct label (Modus Ponens, Modus Tollens, Fallacy of the Converse, Fallacy of the Inverse). 4. Justify your answer in one sentence. Developing Mastery Worksheet Cases 1. Case of the Rainy Day ☔ If it rains, then the ground will be wet. It rains. Therefore, the ground is wet. ➞ Reasoning Type: _____ 2. Case of the Failed Exam 📖 If a student studies hard, then she will pass the	Activity: “Logic Detectives” 🔍 Strategy 1: Role-Play Game (Group Activity) Purpose: To encourage collaboration and competitive learning while identifying argument forms. Materials: • Case cards (or slides) • paper/mini-whiteboards • marker • scoreboard. Instructions: 1. Divide the class into small groups (3–4 members each). 2. Present one case at a time. Groups discuss and write their answers on their mini-board. 3. At the teacher’s signal (“Reveal!”), all groups show their answers. 4. Teacher awards points for correct identification + justification. 5. The group with the highest points wins. Answer Key 1. Case of the Rainy Day ☔ Answer: Modus Ponens ☒ Explanation: If $p \rightarrow q$ and p is true, then q must be true. 2. Case of the Failed Exam 📖 Answer: Modus Tollens ☒ Explanation: If $p \rightarrow q$ and q is false, then p must also be false.

test.
She did not pass the test.
Therefore, she did not study hard.
☞ Reasoning Type: _____

3. **Case of the Traveler** ✈

If I am in Cebu, then I am in the Philippines.
I am in the Philippines.
Therefore, I am in Cebu.
☞ Reasoning Type: _____

4. **Case of the Musician** 🎹

If I practice piano daily, then I will become a good pianist.
I did not practice piano daily.
Therefore, I will not become a good pianist.
☞ Reasoning Type: _____

5. **Case of the Athlete** 🏃

If I exercise regularly, then I will be physically fit.
I am physically fit.
Therefore, I exercise regularly.
☞ Reasoning Type: _____

6. **Case of the Pet Lover** 🐶

If someone owns a dog, then they love animals.
This person does not own a dog.
Therefore, they do not love animals.
☞ Reasoning Type: _____

7. **Case of the Light Bulb** 💡

If the switch is on, then the light will shine.
The light is not shining.
Therefore, the switch is not on.
☞ Reasoning Type: _____

3. **Case of the Traveler** ✈

Answer: Fallacy of the Converse ✕

Explanation: From $p \rightarrow q$ and q , you cannot conclude p . (Being in the Philippines doesn't mean being in Cebu.)

4. **Case of the Musician** 🎹

Answer: Fallacy of the Inverse ✕

Explanation: From $p \rightarrow q$ and $\sim p$, you cannot conclude $\sim q$. (Not practicing daily doesn't automatically mean not becoming a pianist.)

5. **Case of the Athlete** 🏃

Answer: Fallacy of the Converse ✕

Explanation: From $p \rightarrow q$ and q , you cannot conclude p . (Being fit doesn't always mean exercising regularly.)

6. **Case of the Pet Lover** 🐶

Answer: Fallacy of the Inverse ✕

Explanation: From $p \rightarrow q$ and $\sim p$, you cannot conclude $\sim q$. (Not owning a dog doesn't mean not loving animals.)

7. **Case of the Light Bulb** 💡

Answer: Modus Tollens ☑

Explanation: If $p \rightarrow q$ and q is false, then p must be false. (If the light isn't shining, the switch is not on.)

Strategy 2: Movement-Based Quiz (Kinesthetic Activity)

Purpose: To make learning active by letting students physically engage in identifying reasoning types.

Guide Questions:

1. Among all the cases, which reasoning pattern do you think is the most common in everyday situations? Why do you think so?
2. How can identifying valid and invalid reasoning help you make better decisions in real life?
3. Choose one case and change it so that it becomes a **valid** reasoning pattern. Explain your revision.
4. In social media or news posts, do you think people often use **fallacious reasoning**? Give an example and explain what kind of fallacy it might be.
5. Reflect on your experiences in school or at home. When have you or someone else used reasoning similar to **Modus Tollens**? What was the situation and the conclusion?
6. Why do you think it's important for students to learn about logical reasoning patterns like **Modus Ponens** and **Modus Tollens**?

Materials:

- ✓ Four labeled signs posted in the classroom corners: *Modus Ponens*, *Modus Tollens*, *Fallacy of the Converse*, *Fallacy of the Inverse*.

Instructions:

1. Read a case aloud.

2. Students move to the corner representing their answer.

3. Once settled, randomly call on students to justify their choice.

4. Give immediate feedback and explain the correct reasoning.

5. Repeat with all cases until mastery is shown.

Summary of Answer Key:

Case	Reasoning Type
Rainy Day	Modus Ponens ☒
Failed Exam	Modus Tollens ☒
Traveler	Fallacy of the Converse ✗
Musician	Fallacy of the Inverse ✗
Athlete	Fallacy of the Converse ✗
Pet Lover	Fallacy of the Inverse ✗
Light Bulb	Modus Tollens ☒

Rubric for Assessment for the two strategies:

Criteria	4 – Excellent	3 – Proficient	2-Developing	1-Beginning
Accuracy (Correct identification of	Identifies all cases correctly	Identifies 5–6 correctly	Identifies 3–4 correctly	Identifies 0–2 correctly

	Rubric for Assessment:<table><tr><th>Criteria</th><th>4 – Excellent</th><th>3 – Proficient</th><th>2- Developing</th><th>1-Beginning</th></tr><tr><td>Accuracy (Correct identification of reasoning type)</td><td>Identifies all cases correctly</td><td>Identifies 5–6 correctly</td><td>Identifies 3–4 correctly</td><td>Identifies 0–2 correctly</td></tr><tr><td>Explanation/Justification</td><td>Gives clear, logical explanations for all answers</td><td>Gives clear explanations for most answers</td><td>Gives partial or unclear explanations</td><td>Gives little or no explanation</td></tr><tr><td>Participation/Engagement</td><td>Actively participates, collaborates well, and shows enthusiasm</td><td>Participates with some prompting</td><td>Limited participation, needs encouragement</td><td>Passive, little to no participation</td></tr></table> Scoring Guide: 10–12 points = Excellent Mastery 7–9 points = Proficient 4–6 points = Developing 1–3 points = Beginning					Criteria	4 – Excellent	3 – Proficient	2- Developing	1-Beginning	Accuracy (Correct identification of reasoning type)	Identifies all cases correctly	Identifies 5–6 correctly	Identifies 3–4 correctly	Identifies 0–2 correctly	Explanation/Justification	Gives clear, logical explanations for all answers	Gives clear explanations for most answers	Gives partial or unclear explanations	Gives little or no explanation	Participation/Engagement	Actively participates, collaborates well, and shows enthusiasm	Participates with some prompting	Limited participation, needs encouragement	Passive, little to no participation	<table><tr><td>reasoning type)</td><td></td><td></td><td></td><td></td></tr><tr><td>Explanation/Justification</td><td>Gives clear, logical explanations for all answers</td><td>Gives clear explanations for most answers</td><td>Gives partial or unclear explanations</td><td>Gives little or no explanation</td></tr><tr><td>Participation/Engagement</td><td>Actively participates, collaborates well, and shows enthusiasm</td><td>Participates with some prompting</td><td>Limited participation, needs encouragement</td><td>Passive, little to no participation</td></tr></table>	reasoning type)					Explanation/Justification	Gives clear, logical explanations for all answers	Gives clear explanations for most answers	Gives partial or unclear explanations	Gives little or no explanation	Participation/Engagement	Actively participates, collaborates well, and shows enthusiasm	Participates with some prompting	Limited participation, needs encouragement	Passive, little to no participation
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C. Demonstrating Knowledge and Skills	C.1. Finding Practical Application																																								
	Activity C.1: “Spot the Logic!” 🧠 You will identify examples of valid and fallacious reasoning commonly seen in everyday communication, such as social media posts, news reports, or conversations. Classify each example according to its		Activity C.1: “Spot the Logic!” 🧠 Students will identify examples of valid and fallacious reasoning commonly seen in everyday communication, such as social media posts, news reports, or conversations. Classify each example according to its																																						

	type: Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse. Instructions: 1. Observe statements, posts, or conversations related to natural disasters (e.g., typhoons, earthquakes, floods) and mental health awareness (e.g., self-care, seeking help, emotional well-being). 2. Identify two examples of valid reasoning (Modus Ponens or Modus Tollens) and two examples of fallacious reasoning (Fallacy of the Inverse or Fallacy of the Converse). 3. Present your findings using one of the following formats: • Written Output: A worksheet or poster showing: ○ The actual statement or situation ○ The type of reasoning it represents ○ Whether it is valid or a fallacy • Interactive Output: A short role-play or dialogue (2–3 minutes) where you demonstrate each reasoning type in real-life contexts related to natural disasters or mental health. Examples A. Valid Reasoning (Modus Ponens) Situation: During a typhoon alert Ana: If PAGASA declares Signal No. 3, classes are automatically suspended, right? Ben: Yes, they just announced Signal No. 3 in our area. Ana: Then that means there’s no school tomorrow! B. Valid Reasoning (Modus Tollens) Situation: Talking about self-care Mia: If you meditate daily, you’ll feel more relaxed.	type: Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse. Purpose To help students apply logical reasoning to real-life situations, particularly those related to natural disasters and mental health awareness. Through identifying and analyzing examples of valid and fallacious reasoning Materials ✓ Copies of the activity sheet or manila paper for poster output ✓ Markers or coloring materials ✓ Optional: props or cue cards for role-play presentations Instructions: Introduction (5–10 minutes): • Begin with a short review of <i>Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse</i>. • Ask students to share familiar statements they’ve heard related to natural disasters or mental health. • Highlight how logic can be applied in interpreting such statements. Activity Explanation (5 minutes): • Present Activity C.1 “Spot the Logic!” on the board or screen. • Explain that students will identify <i>four examples</i> of reasoning—two valid and two fallacious—from real-life or media-based contexts.
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Jay: Hmm... but I've been feeling anxious lately.
Mia: Then maybe you haven't been meditating regularly.

C. Fallacy of the Inverse

Situation: After an earthquake drill

Leo: If there's an earthquake, we should evacuate immediately.

Kim: There was no earthquake today, so I guess we don't need to practice evacuation.

Leo: That's not right! We still need to practice even without an earthquake.

D. Fallacy of the Converse

Situation: Mental health awareness week

Sam: If a person talks to a counselor, it means they're taking care of their mental health.

Ella: Oh, so if someone takes care of their mental health, they must be seeing a counselor?

Sam: Not necessarily! There are many ways to care for your mental health.

Criteria	4- Excellent	3-Good	2- Fair	1-Needs Improvement
Accuracy of Reasoning Type	Correctly identifies and classifies all reasoning types	Minor errors in classification	Some incorrect classifications	Misclassifies most examples
Relevance to Topic	All examples clearly relate to natural disasters or mental health	Mostly relevant examples	Somewhat related	Unrelated examples
Creativity / Presentation	Engaging and well-structured presentation or output	Clear and organized	Somewhat unclear	Disorganized or incomplete

- Emphasize that the contexts should focus on **natural disasters** and **mental health awareness**.

Group Work / Individual Task (20–25 minutes):

- Students may work **individually or in pairs**.
- Instruct them to:
 - Find or create statements that show logical reasoning.
 - Classify each according to its reasoning type.
 - Indicate whether the reasoning is valid or a fallacy.
 - Present the examples either as:
 - A **written output** (worksheet/poster), or
 - An **interactive output** (short dialogue or role-play).

Presentation and Discussion (15–20 minutes):

- Allow students to share their work or perform their dialogues.
- After each presentation, discuss as a class:
 - Which type of reasoning was demonstrated?
 - Why is it considered valid or fallacious?
 - How does understanding logic help us evaluate real-life situations or media messages?

Wrap-Up (5 minutes):

- Summarize the key points: logic helps us think clearly, avoid faulty conclusions, and make sound judgments.
- Reinforce the importance of applying reasoning skills in interpreting issues like natural disaster responses and mental health decisions.

	Understanding of Valid vs. Fallacious Reasoning	Demonstrates deep understanding	Demonstrates adequate understanding	Limited understanding	Shows confusion or misunderstanding	EXAMPLES: A. Valid Reasoning (Modus Ponens) Situation: During a typhoon alert Ana: If PAGASA declares Signal No. 3, classes are automatically suspended, right? Ben: Yes, they just announced Signal No. 3 in our area. Ana: Then that means there's no school tomorrow! <i>Type of Reasoning: Modus Ponens</i> <i>Explanation:</i> The reasoning follows logically — if the condition happens (Signal No. 3), then suspension follows. B. Valid Reasoning (Modus Tollens) Situation: Talking about self-care Mia: If you meditate daily, you'll feel more relaxed. Jay: Hmm... but I've been feeling anxious lately. Mia: Then maybe you haven't been meditating regularly. <i>Type of Reasoning: Modus Tollens</i> <i>Explanation:</i> It negates the consequence to deny the cause — valid reasoning. C. Fallacy of the Inverse Situation: After an earthquake drill Leo: If there's an earthquake, we should evacuate immediately. Kim: There was no earthquake today, so I guess we don't need to practice evacuation. Leo: That's not right! We still need to practice even without an earthquake. <i>Type of Reasoning: Fallacy of the Inverse</i> <i>Explanation:</i> The logic is incorrect — the absence of the condition doesn't mean the action is unnecessary.

		D. Fallacy of the Converse Situation: Mental health awareness week Sam: If a person talks to a counselor, it means they’re taking care of their mental health. Ella: Oh, so if someone takes care of their mental health, they must be seeing a counselor? Sam: Not necessarily! There are many ways to care for your mental health. <i>Type of Reasoning: Fallacy of the Converse</i> <i>Explanation:</i> The statement reverses the cause and effect — caring for mental health doesn’t always mean counseling.Rubric for Assessment: <table><tr><th>Criteria</th><th>4– Excellent</th><th>3-Good</th><th>2- Fair</th><th>1-Needs Improve ment</th></tr><tr><td>Accuracy of Reasoning Type</td><td>Correctly identifies and classifies all reasoning types</td><td>Minor errors in classification</td><td>Some incorrect classifications</td><td>Misclassifies most examples</td></tr><tr><td>Relevance to Topic</td><td>All examples clearly relate to natural disasters or mental health</td><td>Mostly relevant examples</td><td>Somewhat related</td><td>Unrelated examples</td></tr><tr><td>Creativity / Presentation</td><td>Engaging and well-structured presentation or output</td><td>Clear and organized</td><td>Somewhat unclear</td><td>Disorganized or incomplete</td></tr><tr><td>Understanding of Valid vs. Fallacious Reasoning</td><td>Demonstrates deep understanding</td><td>Demonstrates adequate understanding</td><td>Limited understanding</td><td>Shows confusion or misunderstanding</td></tr></table>	Criteria	4– Excellent	3-Good	2- Fair	1-Needs Improve ment	Accuracy of Reasoning Type	Correctly identifies and classifies all reasoning types	Minor errors in classification	Some incorrect classifications	Misclassifies most examples	Relevance to Topic	All examples clearly relate to natural disasters or mental health	Mostly relevant examples	Somewhat related	Unrelated examples	Creativity / Presentation	Engaging and well-structured presentation or output	Clear and organized	Somewhat unclear	Disorganized or incomplete	Understanding of Valid vs. Fallacious Reasoning	Demonstrates deep understanding	Demonstrates adequate understanding	Limited understanding	Shows confusion or misunderstanding
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C.2. Making Generalization

Activity C.2: Wrapping Up the Lesson

✱ Mnemonic: “PoTo InCo” ✱

- **Po = Modus Ponens** (VALID ☒) → affirms the first (If $p \rightarrow q$, p true $\rightarrow q$ true)
- **To = Modus Tollens** (VALID ☒) → denies the last (If $p \rightarrow q$, q false $\rightarrow p$ false)
- **In = Fallacy of the Inverse** (INVALID ☒) → wrongly flips the “not”
- **Co = Fallacy of the Converse** (INVALID ☒) → wrongly flips the order

Memory Aid

☞ “PoTo is VALID, InCo is INVALID.”

- If it looks like **Po** or **To** → it’s **valid reasoning**.
- If it looks like **In** or **Co** → it’s a **fallacy**.

Activity C.2: Wrapping Up the Lesson

✱ Mnemonic: “PoTo InCo” ✱

Purpose

- To help students consolidate their understanding of **Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse** by arriving at a clear and memorable summary.
- To strengthen critical thinking by guiding students to see the **difference between valid reasoning and fallacies**.

Materials

- ✓ Whiteboard/blackboard or projector
- ✓ Marker/chalk
- ✓ Student notes/notebooks
- ✓ (Optional) Visual aid/anchor chart of the mnemonic “PoTo InCo”

Instructions:

1. Review

- Briefly recap the four reasoning types by asking students: “Who remembers what Modus Ponens is? Modus Tollens?”
- Accept quick answers and write them on the board.

2. Guided Discussion

- Ask students: “Looking at all these reasoning forms, what pattern do you notice

		<i>between Modus Ponens & Modus Tollens?”</i> → Guide them to see that both are valid. ○ Then ask: “<i>What about the Fallacy of the Inverse and Converse?</i>” → Guide them to see that both are invalid fallacies. 3. Generalization Construction ○ Together with the class, write a short generalization statement on the board. Example: <i>“Valid reasoning follows rules (Modus Ponens & Modus Tollens), while faulty reasoning comes from mixing up the order or negation (Fallacy of Inverse & Converse).”</i> 4. Memory Aid (Optional) ○ Introduce the mnemonic “PoTo InCo” → <i>“PoTo is VALID, InCo is INVALID.”</i> ○ Let students chant or repeat it for easier recall. 5. Note-taking ○ Instruct students to copy the final generalization in their notebooks. Encourage them to underline or highlight the terms Valid vs Invalid.
	<i>C.3. Evaluating Learning</i>	
	Activity C.3: Assessing Understanding Formative Assessment	Activity C.3: Assessing Understanding Formative Assessment

Part A: Identification

Tell whether the reasoning is Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse. (1 point each)

1. If it is Saturday, then we don't have class. It is Saturday. Therefore, we don't have class.
2. If a student studies hard, then they will pass the test. The student did not pass the test. Therefore, they did not study hard.
3. If an animal is a dog, then it is a mammal. This animal is a mammal. Therefore, it is a dog.
4. If a person is a teacher, then they can speak in class. This person is not a teacher. Therefore, they cannot talk in class.
5. If the lights are on, then someone is home. The lights are not on. Therefore, no one is home.

Part B: Application

Write your own example (1 each) for:

1. Modus Ponens (valid)
2. Modus Tollens (valid)
3. Fallacy of the Inverse (invalid)
4. Fallacy of the Converse (invalid)

Part C. Synthesis:

1. *How did you arrive at your answers?*

2. *How did you find the activity? Can you share and describe your experience?*

3. *How did understanding different forms of reasoning (Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse) make it easier to*

Purpose

To evaluate students' understanding of the four reasoning types: **Modus Ponens, Modus Tollens, Fallacy of the Inverse, and Fallacy of the Converse.**

To check if students can **identify, differentiate, and apply** these logical structures in both given and self-made examples.

Materials

- ✓ Printed quiz sheet / projector display of the activity
- ✓ Answer sheets or notebooks
- ✓ Pen/pencil

Instructions:

1. Preparation

- Distribute the quiz sheets (or present the items on the board/projector).
- Remind students of the four reasoning forms and give them 2–3 sample items as review.

2. Part A: Identification

- Ask students to read each case carefully.
- Instruct them to determine whether the reasoning follows *Modus Ponens, Modus Tollens, Fallacy of the Inverse, or Fallacy of the Converse.*

3. Part B: Application

- Require students to write **original examples** for each reasoning type.
- Encourage them to connect examples to real-life, school situations, or simple daily routines (e.g., "If I wake up late, then I miss the bus").

4. Checking & Feedback

- Check Part A using the answer key.

	<i>analyze real-life statements or situations? Can you think of a real-life scenario where applying correct reasoning would be helpful?</i> 4. <i>What's the most important thing you learned from this activity about identifying valid and fallacious reasoning? How can this help you make better judgments in real-life situations?</i>	○ For Part B, accept varied responses as long as they match the reasoning type. ○ Provide quick oral feedback by showing sample correct answers. Answer Key Part A 1. Modus Ponens ☒ 2. Modus Tollens ☒ 3. Fallacy of the Converse ✗ 4. Fallacy of the Inverse ✗ 5. Modus Tollens ☒ Part B (Sample answers – students may vary) 5. Reflection Prompt (Optional) ○ End the quiz by asking: “Which reasoning type is easiest for you to use, and why? Which one do you think people mistakenly use most in daily life?” ○ This helps students connect logical reasoning with critical thinking in real-life decision-making. Part C. (Sample answers – students may vary)
	C.4. Additional Activities Activity C.4: Conclude! Instructions: 1. <i>Read each statement or situation carefully.</i> 2. <i>Identify the conclusion that logically follows based on the given information.</i> 3. <i>Decide whether the reasoning used is Modus Ponens or Modus Tollens.</i>	Activity C.4: Conclude! Purpose: To apply Modus Ponens and Modus Tollens in real-life and logical scenarios through guided reasoning and discussion.

4. Write your complete conclusion in the space provided.
5. Be ready to explain your answer or act it out in class as a short dialogue.

 **Tip:**

- Modus Ponens follows the pattern: If $P \rightarrow Q$; P ; $\therefore Q$

*Example: If it rains, the ground gets wet. It rained.
Therefore, the ground is wet.*

- Modus Tollens follows the pattern: If $P \rightarrow Q$; $\sim Q$; $\therefore \sim P$

Example: If it rains, the ground gets wet. The ground is not wet. Therefore, it did not rain.

*1. If I pass the final exam, I will graduate. I graduated.
Therefore...*

2. I will go to La Union, I will eat "Halo-Halo" de Iloko. If I eat "Halo-Halo", I gain weight. Therefore,...

3. Rachel and Paulo will be at a concert. Rachel was at the concert. Therefore,...

4. If a person is an engineer, then that person has a college degree. Dennis does not have a college degree. Therefore,...

5. If I am irritated or angry, I cannot concentrate. I can concentrate. Therefore,...

Strategy 1: Guided Discovery (Inquiry-Based Approach)

Objective: Help students identify and explain whether each statement follows *Modus Ponens* or *Modus Tollens* through step-by-step reasoning.

Procedure:

1. Begin the activity by recalling the two reasoning patterns:
 - **Modus Ponens:** If $P \rightarrow Q$; P ; $\therefore Q$
 - **Modus Tollens:** If $P \rightarrow Q$; $\sim Q$; $\therefore \sim P$
2. Ask students to **read each scenario aloud** and **analyze** it.
3. Guide them with questions such as:
 - "What is the condition (If...)? What is the result (Then...)? "
 - "Was the condition affirmed or denied?"
4. Encourage students to **write the conclusion** and **name** the type of reasoning used.
5. Facilitate a short class discussion for each item to confirm answers and clarify misconceptions.

Teacher's Note:

 Encourage learners to explain their reasoning in their own words before revealing the logical pattern. This deepens understanding rather than rote memorization.

Strategy 2: Collaborative Conversation (Contextual Approach)

		Objective: Allow students to see how Modus Ponens and Modus Tollens appear in everyday conversations. Instructions: 1. Divide the class into small groups. 2. Assign one or two examples from the activity to each group. 3. Instruct students to rewrite their assigned statements as short dialogues between two people (e.g., friends, family members, or classmates). a. Example: A: If I pass the final exam, I will graduate. B: You passed the final exam, right? A: Yes! Therefore, I will graduate! 4. Let groups present their dialogues to the class. 5. After each presentation, ask: a. “Which rule of inference did they use — Modus Ponens or Modus Tollens?” b. “How does this type of reasoning apply to real life?” Teacher’s Note:  Turning the logical statements into conversations makes learning interactive, fun, and relatable. It also helps students see logic as part of everyday reasoning
VI. SUMMATIVE ASSESSMENTS	Please refer to the attachments for the summative assessment and performance task.	

VII. REFLECTION

Reflection is most effective when done regularly. Make it a habit to reflect after each lesson or weekly to track your progress and growth over time. The reflection questions are designed to help you thoughtfully evaluate your teaching, identify strengths, recognize challenges, and plan improvements to enhance learner learning. The questions provided below may help in reflecting to your teaching performance.

1. What aspects of your teaching went particularly well?

- Which part of the lesson engaged learners the most?
- What teaching strategies or methods contributed to successful learning?
- How did learners respond to your instruction (e.g., participation, performance, behavior)?
- What evidence shows that learning objectives were achieved?
- What personal teaching strengths were evident in this session?

2. What challenges did you encounter, and how did you address them?

- What difficulties arose during lesson delivery or classroom management?
- Were there moments when learners struggled to understand a concept? Why?
- How did you adjust your strategies or pacing to respond to these challenges?
- What support or resources would have helped you handle the situation more effectively?
- What will you do differently next time to overcome similar obstacles?

3 What will you do differently next time to improve learner learning?

- Which teaching strategies or approaches would you modify or try in the future?
- How will you better address diverse learning needs in your classroom?
- Are there new resources, activities, or technologies you plan to incorporate?
- What steps can you take to enhance learner engagement or participation?
- How will you assess whether these changes improve learning outcomes?

UNIT TEST
Unit Topics: Propositions, Conditionals, and Syllogisms

Name: _____
Grade/Section: _____

Date: _____
Score: _____ / 25

Directions: Read each item carefully. Choose the letter of the correct answer.

1. Which of the following is a proposition?
A. "What time is it?"
B. "Close the door."
C. "The sun rises in the east."
D. "Hurry up!"
2. Which statement is a **simple proposition**?
A. "It is raining and it is cold."
B. "All triangles have three sides."
C. "If it rains, then the ground is wet."
D. "The bell rings or the class starts."
3. Which is a **compound proposition**?
A. "Manila is the capital of the Philippines."
B. "The Earth orbits the sun."
C. " $2 + 2 = 4$."
D. "Baguio is cold."
4. What is the **negation** of "The sky is blue"?
A. The sky is not blue.
B. The sky might be blue.
C. If the sky is blue, then it is day.
D. The sky is blue or cloudy.
5. Which of the following illustrates a **conjunction ($\wedge$)**?
A. "He is tall or he is strong."
B. "If she studies, she will pass."
C. "It is raining and it is windy."
D. "The cake is not sweet."
6. Which is an example of a **disjunction ($\vee$)**?
A. "The book is old and torn."
B. "The dog is not barking."
C. "The student is diligent or intelligent."
D. "The teacher is kind."
7. Which of the following is NOT a proposition?
A. "The classroom is noisy."
B. "Cebu is in the Visayas."
C. "5 is a prime number."
D. "Do you like ice cream?"
8. Which proposition is **True**?
A. "The Pacific Ocean is smaller than the Atlantic Ocean."
B. "The Earth has one moon."

- C. "2 is greater than 5."
D. "All squares are circles."
9. "If the computer is on, then the screen has light." This is best described as:
A. Negation
B. Conjunction
C. Conditional
D. Disjunction
10. "The teacher is kind and patient." This statement is:
A. Simple proposition
B. Compound proposition
C. Negation
D. Fallacy
11. Which of the following is a **conditional proposition**?
A. "It rains and I bring an umbrella."
B. "It does not rain."
C. "It rains or I bring an umbrella."
D. "If it rains, then I bring an umbrella."
12. What is the **converse** of "If it rains, then the ground is wet"?
A. If the ground is not wet, then it does not rain.
B. If it does not rain, then the ground is not wet.
C. If it rains, then the ground is not wet.
D. If the ground is wet, then it rains.
13. What is the **inverse** of "If the car is fast, then it consumes more fuel"?
A. If the car is not fast, then it consumes less fuel.
B. If the car consumes more fuel, then it is fast.
C. If the car is not fast, then it does not consume more fuel.
D. If the car does not consume more fuel, then it is not fast.
14. What is the **contrapositive** of "If a figure is a square, then it is a rectangle"?
A. If it is not a square, then it is not a rectangle.
B. If it is not a rectangle, then it is not a square.
C. If it is a rectangle, then it is a square.
D. If it is a square, then it is not a rectangle.
15. Which of the following is TRUE about conditionals?
A. A converse and contrapositive are always the same.
B. An inverse and converse are logically equivalent.
C. A conditional and its contrapositive are logically equivalent.
D. A conditional and its inverse are always equivalent.
16. Which syllogism follows the rule of **modus ponens**?
A. If it rains, then the ground is wet. It did not rain. Therefore, the ground is not wet.
B. If it rains, then the ground is wet. The ground is not wet. Therefore, it did not rain.
C. If it rains, then the ground is wet. The ground is wet. Therefore, it rained.
D. If it rains, then the ground is wet. It rained. Therefore, the ground is wet.

17. Which syllogism illustrates **modus tollens**?
- A. If she studies, then she passes. She studies. Therefore, she passes.
 - B. If she studies, then she passes. She passes. Therefore, she studied.
 - C. If she studies, then she passes. She did not pass. Therefore, she did not study.**
 - D. If she studies, then she passes. She did not pass. Therefore, she studied.
18. Which of the following is an **inverse fallacy**?
- A. If it rains, then the ground is wet. It did not rain. Therefore, the ground is not wet.**
 - B. If it rains, then the ground is wet. The ground is wet. Therefore, it rained.
 - C. If it rains, then the ground is wet. It rains. Therefore, the ground is wet.
 - D. If it rains, then the ground is wet. The ground is not wet. Therefore, it did not rain.
19. Which of the following is a **converse fallacy**?
- A. If the lamp is on, then the room is bright. The room is bright. Therefore, the lamp is on.**
 - B. If the lamp is on, then the room is bright. The lamp is not on. Therefore, the room is not bright.
 - C. If the lamp is on, then the room is bright. The lamp is on. Therefore, the room is bright.
 - D. If the lamp is on, then the room is bright. The room is not bright. Therefore, the lamp is not on.
20. Which of the following is a **tautology**?
- A. "It will rain or it will not rain."**
 - B. "The square is a circle."
 - C. "If it rains, then the ground is dry."
 - D. "The sun is hot and the moon is hot."
21. Which is a **contradiction**?
- A. "The sky is blue or the sky is not blue."
 - B. "All bachelors are unmarried."
 - C. "The triangle has four sides."**
 - D. "The Earth orbits the sun."
22. Which of the following illustrates a **valid syllogism**?
- A. Some birds can fly. Penguins can't fly. Therefore, penguins are not birds.
 - B. All dogs bark. Cats bark. Therefore, cats are dogs.
 - C. If today is Sunday, then tomorrow is Monday. Tomorrow is Monday. Therefore, today is Sunday.
 - D. All men are mortal. Socrates is a man. Therefore, Socrates is mortal.**
23. Which is an example of a **fallacy of assumption**?
- A. All flowers need sunlight. This plant needs sunlight. Therefore, this plant is a flower.**
 - B. If it rains, the street gets wet. It rained. Therefore, the street is wet.
 - C. If it rains, the street gets wet. The street is not wet. Therefore, it did not rain.
 - D. Either the light is on or it is off. The light is on. Therefore, it is not off.

24. Which of the following best describes a **logical argument**?
- A. A statement that is always true regardless of truth values.
 - B. A set of statements where a conclusion follows from premises.**
 - C. A compound proposition with only negations.
 - D. An emotional appeal to persuade others.
25. Which of the following is an example of **practical application of logic**?
- A. Deciding based on gossip.
 - B. Relying only on popularity to decide.
 - C. Choosing based on emotions only.
 - D. Using facts to support a debate argument.**
26. If it rains, the ground becomes wet. It is raining. Which conclusion is logically correct?
- A. The ground is wet.**
 - B. The ground is dry.
 - C. It is not raining.
 - D. The rain stopped.
27. If I study hard, I will pass the test. I did not pass the test. What can be concluded?
- A. I studied hard.
 - B. I did not study hard.**
 - C. The test was difficult.
 - D. I forgot to review.
28. If a person lives in Baguio, then they live in the Philippines.
Maria lives in the Philippines.
Which reasoning error is shown below?
- A. Modus Ponens
 - B. Converse Fallacy**
 - C. Modus Tollens
 - D. Inverse Fallacy
29. If a number is divisible by 4, then it is even.
This number is not divisible by 4.
Which incorrect conclusion follows an inverse fallacy?
- A. The number is even.
 - B. The number is divisible by 2.
 - C. The number is prime
 - D. The number is not even.**
30. If the alarm is set, then the house is secure.
The house is not secure.
What must be true?
- A. The alarm was not set.**
 - B. The alarm is still ringing.
 - C. The alarm was set late.
 - D. The alarm does not work.
31. If a student is diligent, then they pass the course.
Rico is diligent.
Which conclusion is valid?

- A. Rico did not attend class.
- B. Rico failed the course.
- C. Rico passed the course.**
- D. Rico was absent often.

32. If the light is on, then there is electricity.

There is electricity.

Which reasoning mistake is made if someone concludes that *the light is on*?

- A. Modus Ponens
- B. Converse Fallacy**
- C. Modus Tollens
- D. Inverse Fallacy.

33. If a device is fully charged, then it can last for 10 hours.

It lasted for 10 hours.

Which fallacy occurs if someone assumes the device was fully charged?

- A. Modus Ponens
- B. Modus Tollens
- C. Converse Fallacy**
- D. Inverse Fallacy

34. If I am in Cebu, then I am in the Philippines.

I am not in Cebu. Therefore, I am not in the Philippines.

- A. Modus Ponens
- B. Modus Tollens
- C. Converse Fallacy
- D. Inverse Fallacy**

35. Anna: If I study hard, I'll pass the exam.

Ben: You didn't pass the exam, so you probably didn't study hard.

What reasoning is used?

- A. Modus Tollens**
- B. Modus Ponens
- C. Converse Fallacy
- D. Inverse Fallacy

Unit 12 - Performance Task

Title: Logic in Action – Role-Playing Real-World Arguments

In this task, you and your group will perform a short role-play that shows a real-life situation (such as a debate, advertisement, sports commentary, or business pitch). After the performance, you will analyze the arguments by making inferences and explaining your reasoning. You will also evaluate whether the arguments are valid, sound, or fallacious. This activity will help you see how logic is used in everyday life to make clear decisions and avoid faulty conclusions.

G – Goal

Your goal is to demonstrate how logic operates in real life by performing and analyzing a short scenario that involves reasoning and persuasion. You will show how to make inferences and evaluate arguments as valid, sound, or fallacious, applying logical principles to real-world contexts such as politics, business, sports, or social media.

R – Role

You are both a role-player and a logic analyst. As a performer, you will act out a realistic situation involving arguments or persuasion. As an analyst, you will step out of character afterward to explain and evaluate the reasoning used in your role-play.

A – Audience

Your audience will be your classmates and teacher, who will assess how effectively you apply logical reasoning in your performance and post-analysis discussion.

S – Situation

Everyday life is filled with reasoning—some sound, others misleading. Whether in debates, advertisements, sports commentaries, business meetings, or social media discussions, arguments are used to persuade or justify opinions. Your group must present a short 2–3 minute role play that depicts one of these real-life situations and then analyze it to determine the validity and soundness of the reasoning involved.

P – Product/Performance

You will present a Role-Play and Analysis in four parts:

1. *Role-Play (2–3 minutes).* Present a short dramatization of a real-world scenario that includes one or more arguments or persuasive exchanges.

Scenario:

The situation centers on a piece of misleading or fake news that spreads among students, friends, or online users. The characters react differently—some believe and share the information right away, while others question its truth and use logical reasoning and fact-checking to verify it. Through your performance, show how false assumptions, hasty generalizations, or emotional appeals can lead people to accept misinformation, and how applying logical propositions and evidence-based reasoning helps uncover the truth and prevent the spread of fake news.

2. *Inference-Making.* After the role play, identify and explain at least two logical inferences that can be drawn from your scenario, supported with reasoning.
3. *Argument Evaluation.* Identify at least two arguments from your role play and evaluate each as valid, sound, or fallacious, explaining why.

4. *Processing & Reflection*. Share your group's insights on how logic helps in making clear decisions, resolving conflicts, and avoiding faulty conclusions in real-life situations.

S – Standards & Criteria for Success

Your performance will be assessed based on:

Criteria	Outstanding (5)	Very Good (4)	Satisfactory (3)	Needs Improvement (2)	Beginning (1)
Relevance and Realism (20%)	Scenario is highly realistic, strongly reflects a real-world situation, and arguments are authentic and convincing.	Scenario is realistic, reflects real-world situations, and arguments are mostly authentic.	Scenario is somewhat realistic; arguments are present but may lack depth.	Scenario shows little realism; arguments are weak or limited.	Scenario is unrealistic; arguments are missing or irrelevant.
Application of Logic (40%)	Inferences are very clear, strongly justified, and arguments are thoroughly and correctly evaluated (valid/sound/fallacious).	Inferences are clear and justified; arguments are mostly evaluated correctly.	Inferences are understandable but partially justified; argument evaluation is sometimes flawed.	Inferences are weak or unclear; argument evaluation shows frequent errors.	Inferences are absent; argument evaluation is mostly incorrect.
Clarity of Analysis (20%)	Explanations are very well-structured, logical, and easy to follow; ideas flow smoothly and coherently.	Explanations are organized, mostly logical, and generally easy to follow.	Explanations show basic organization; reasoning is sometimes unclear.	Explanations are poorly organized; reasoning is often confusing.	Explanations are disorganized and illogical; difficult to follow.
Creativity and Engagement (10%)	Role-play is highly creative, original, and fully engaging, sustaining audience attention throughout.	Role-play is creative and engaging, holding audience attention most of the time.	Role-play shows some creativity; audience engagement is moderate.	Role-play has minimal creativity; audience engagement is low.	Role-play lacks creativity and fails to engage the audience.
Collaboration (10%)	All group members participate equally and contribute meaningfully in both role-play and analysis.	Most group members participate and contribute meaningfully.	Some group members participate; contribution is uneven.	Few group members participate; contribution is limited.	One or two members do most of the work; others do not participate.